

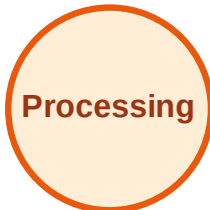
BFS Traversal

Step 1: Start at A

Legend



Complete



Processing



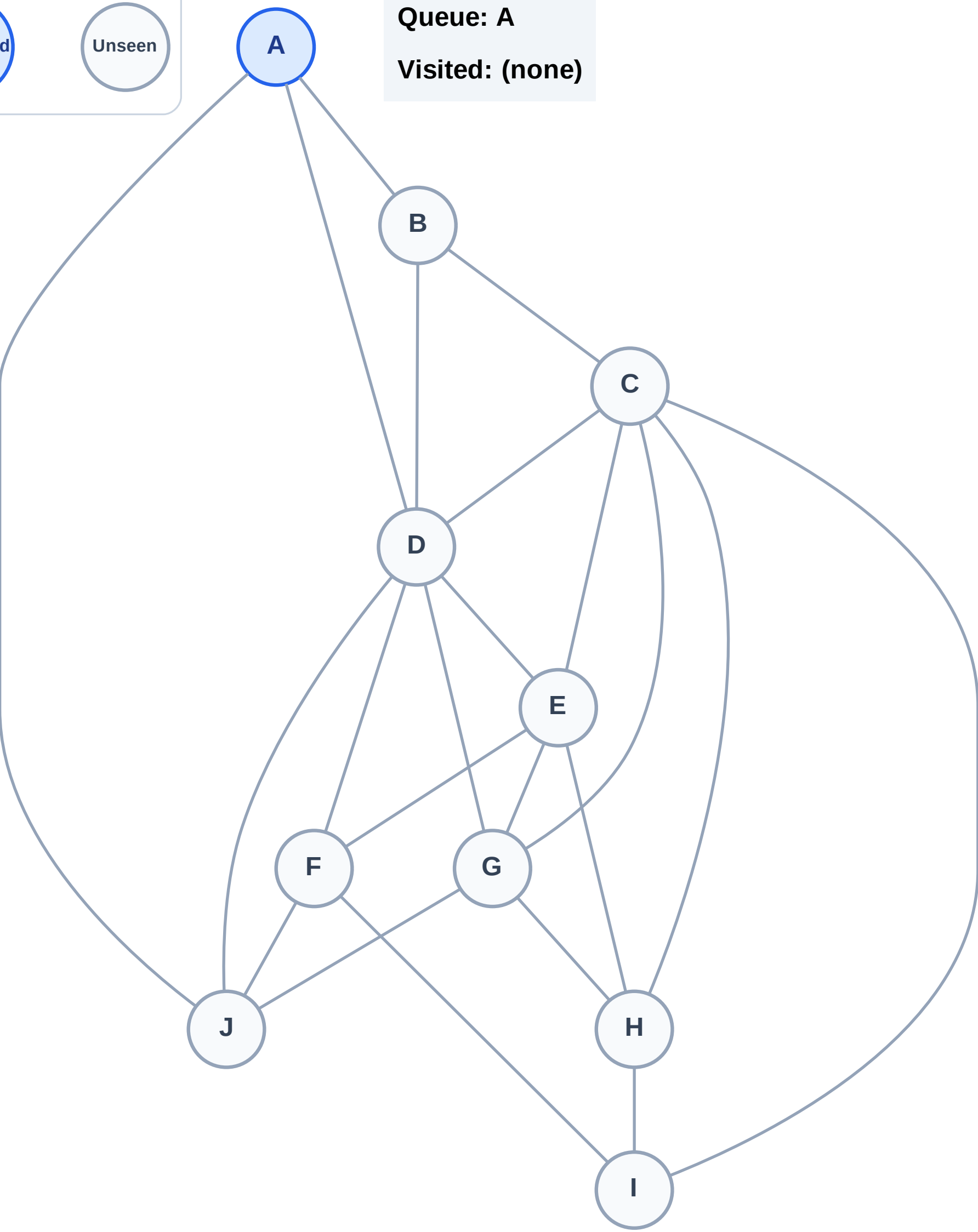
Discovered



Unseen

Queue: A

Visited: (none)



BFS Traversal

Step 2: Process node A

Legend

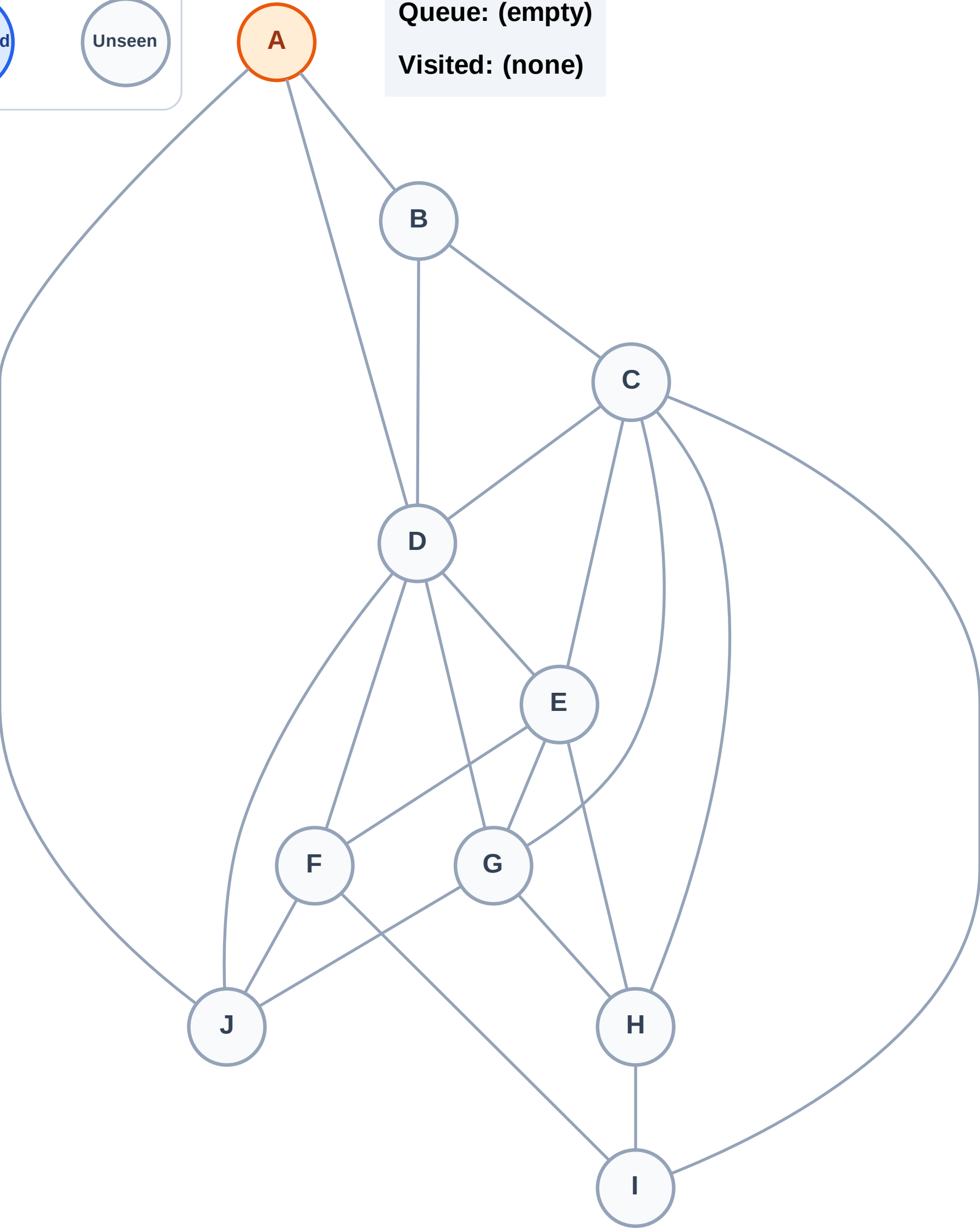
Complete

Processing

Discovered

Unseen

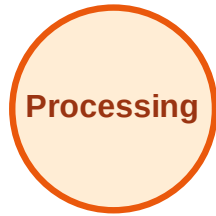
Queue: (empty)
Visited: (none)



BFS Traversal

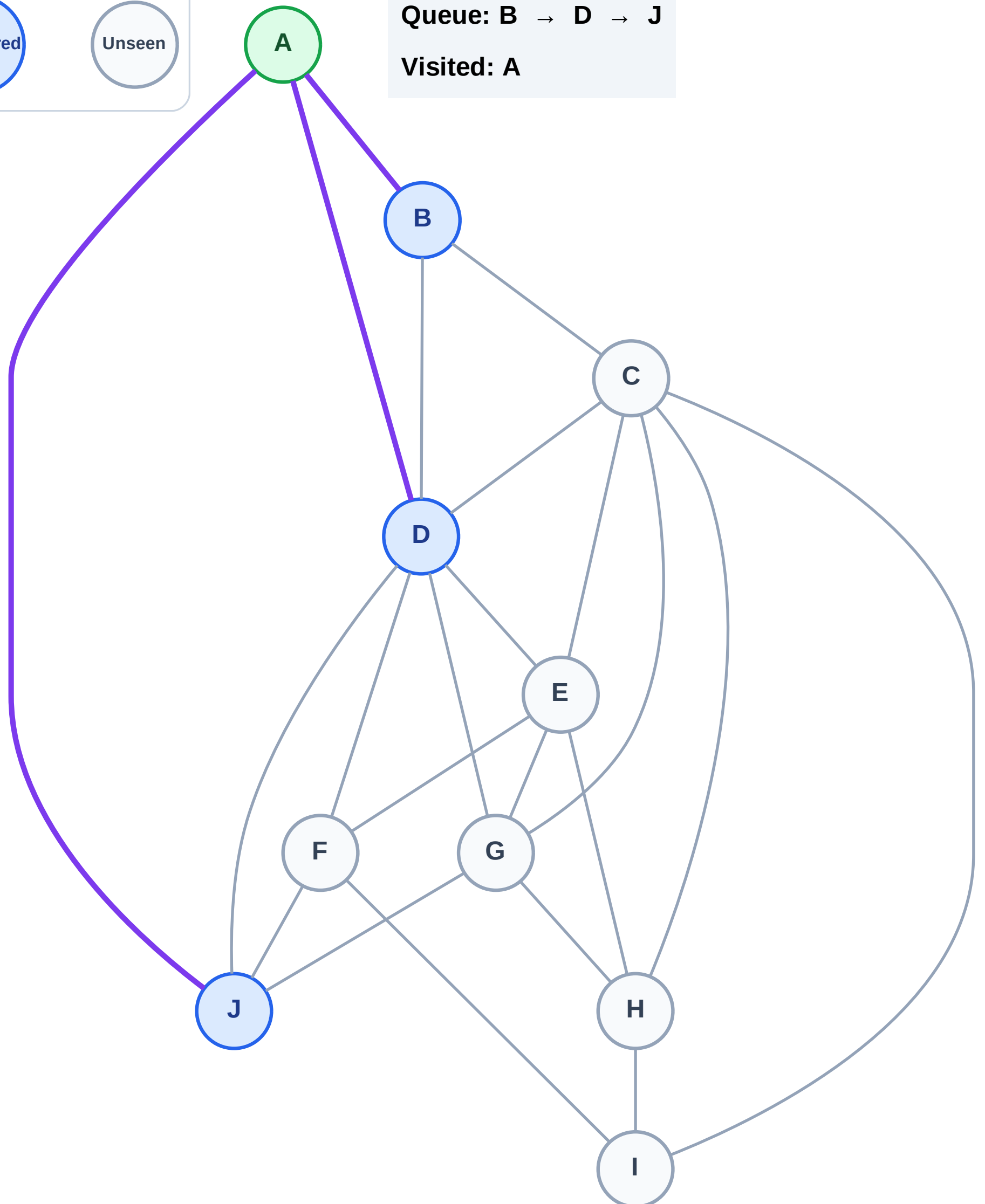
Step 3: Discover B, D, J from A

Legend



Queue: B → D → J

Visited: A



BFS Traversal

Step 4: Process node B

Legend

Complete

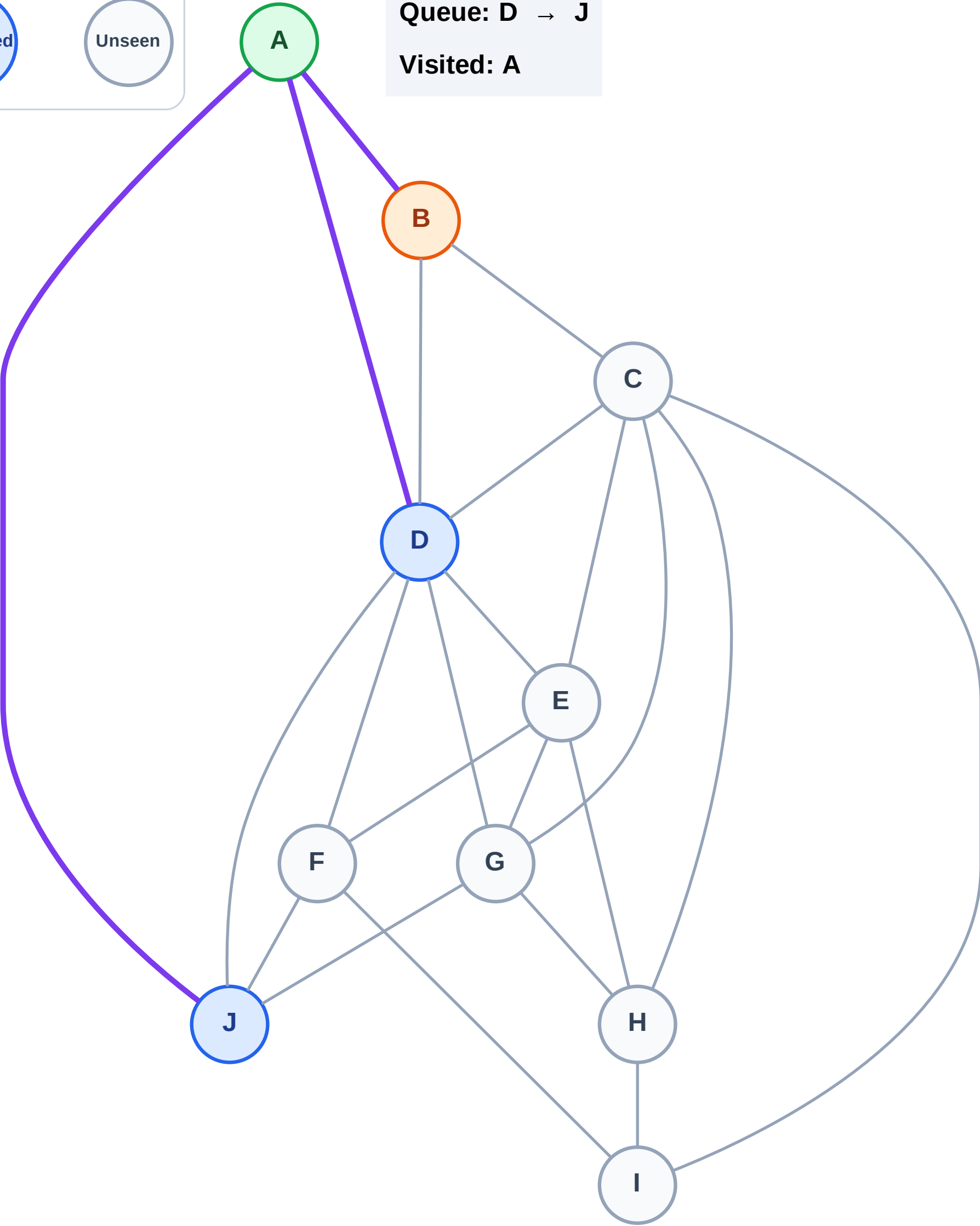
Processing

Discovered

Unseen

Queue: D → J

Visited: A



BFS Traversal

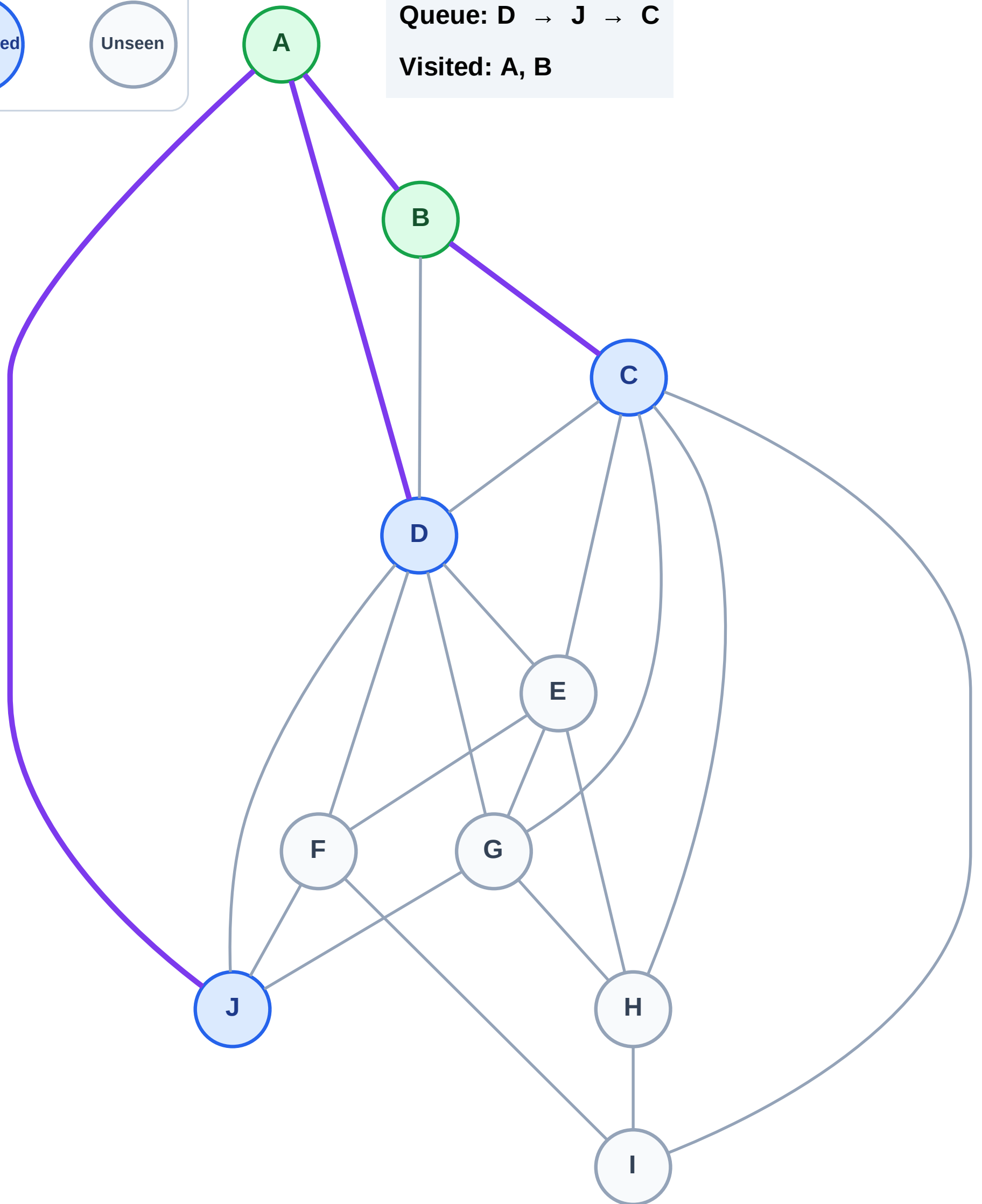
Step 5: Discover C from B

Legend



Queue: D → J → C

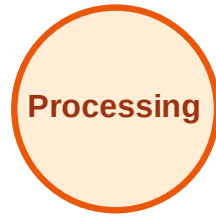
Visited: A, B



BFS Traversal

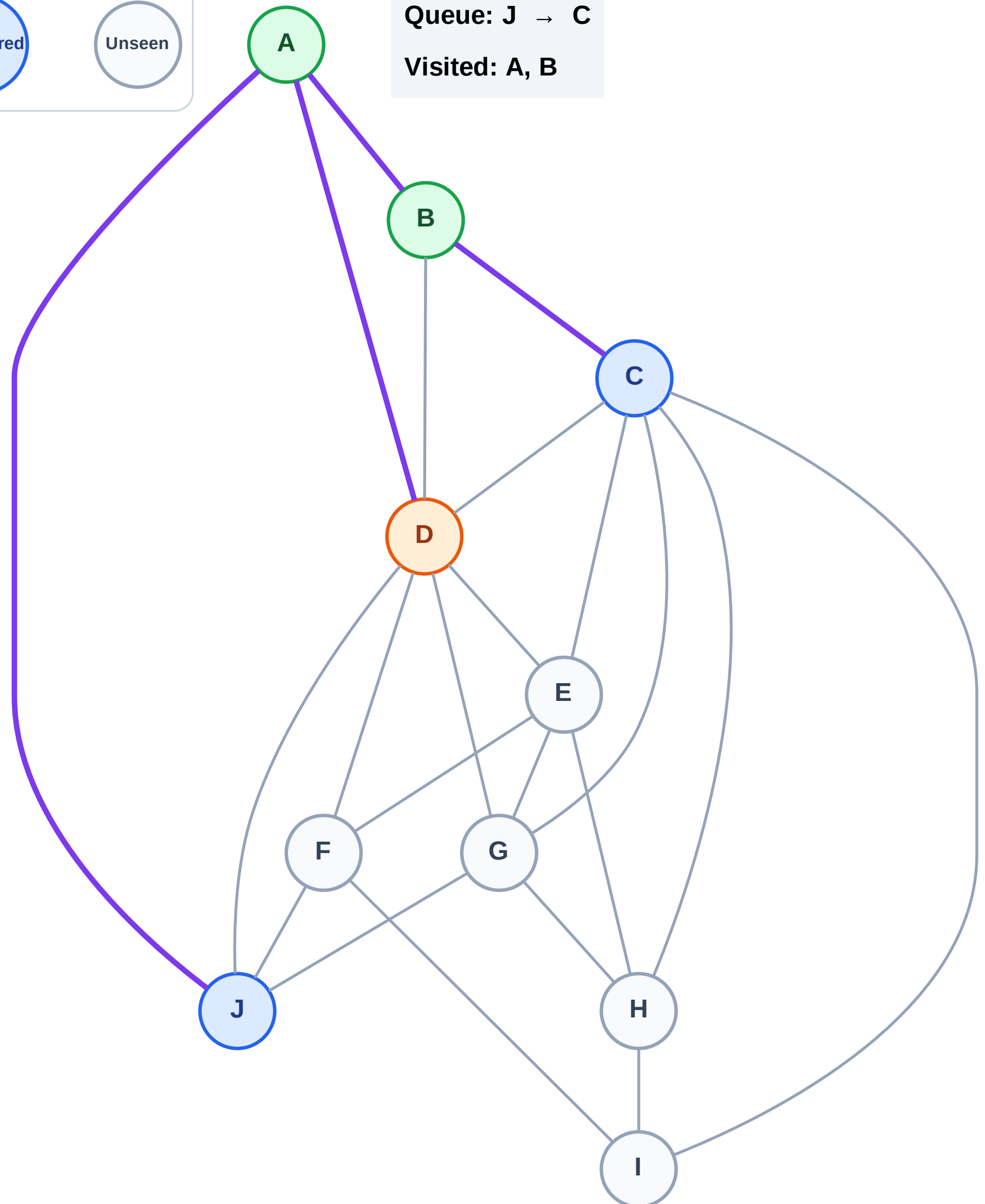
Step 6: Process node D

Legend



Queue: J → C

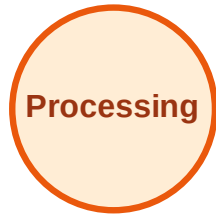
Visited: A, B



BFS Traversal

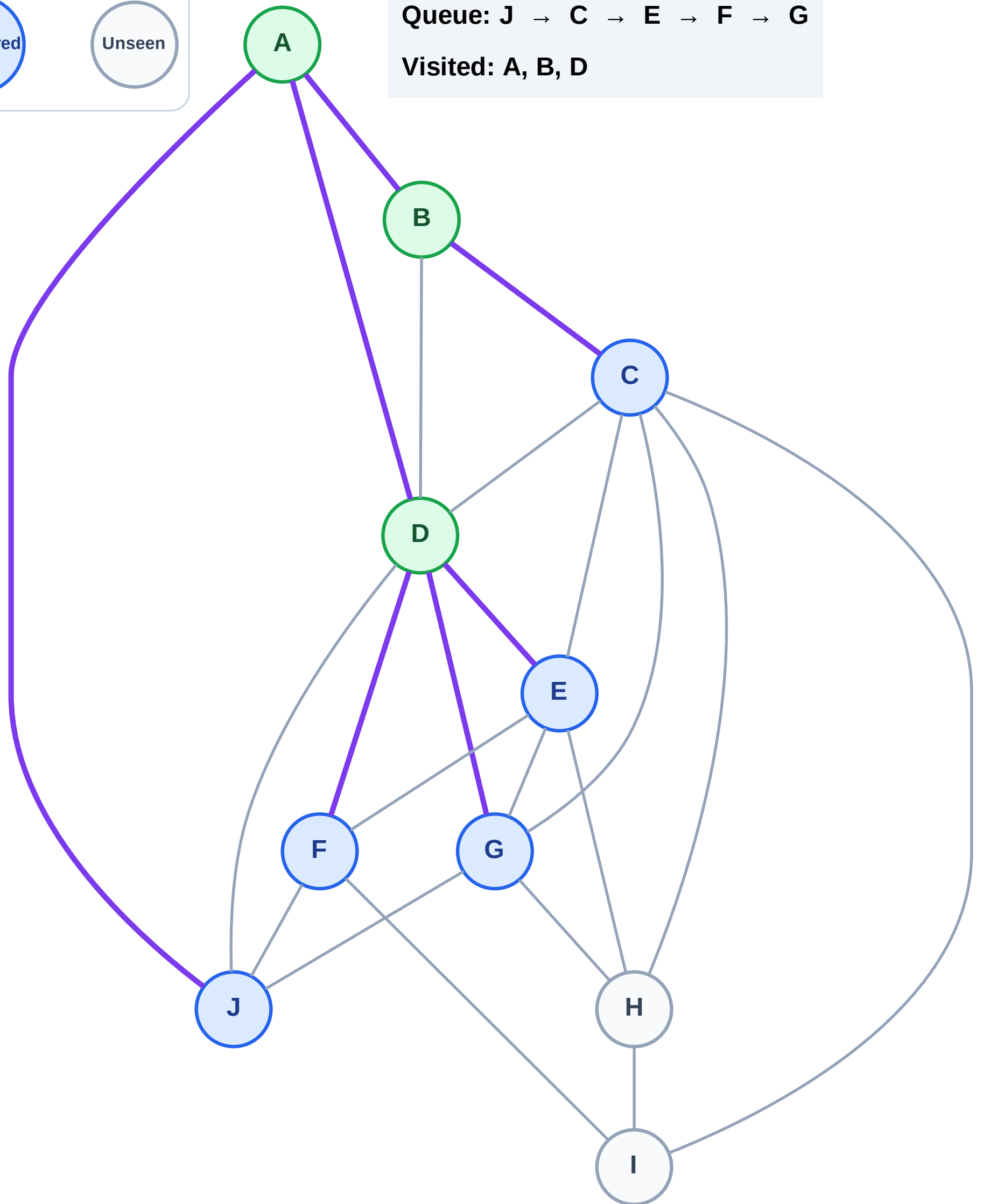
Step 7: Discover E, F, G from D

Legend



Queue: J → C → E → F → G

Visited: A, B, D



BFS Traversal

Step 8: Process node J

Legend

Complete

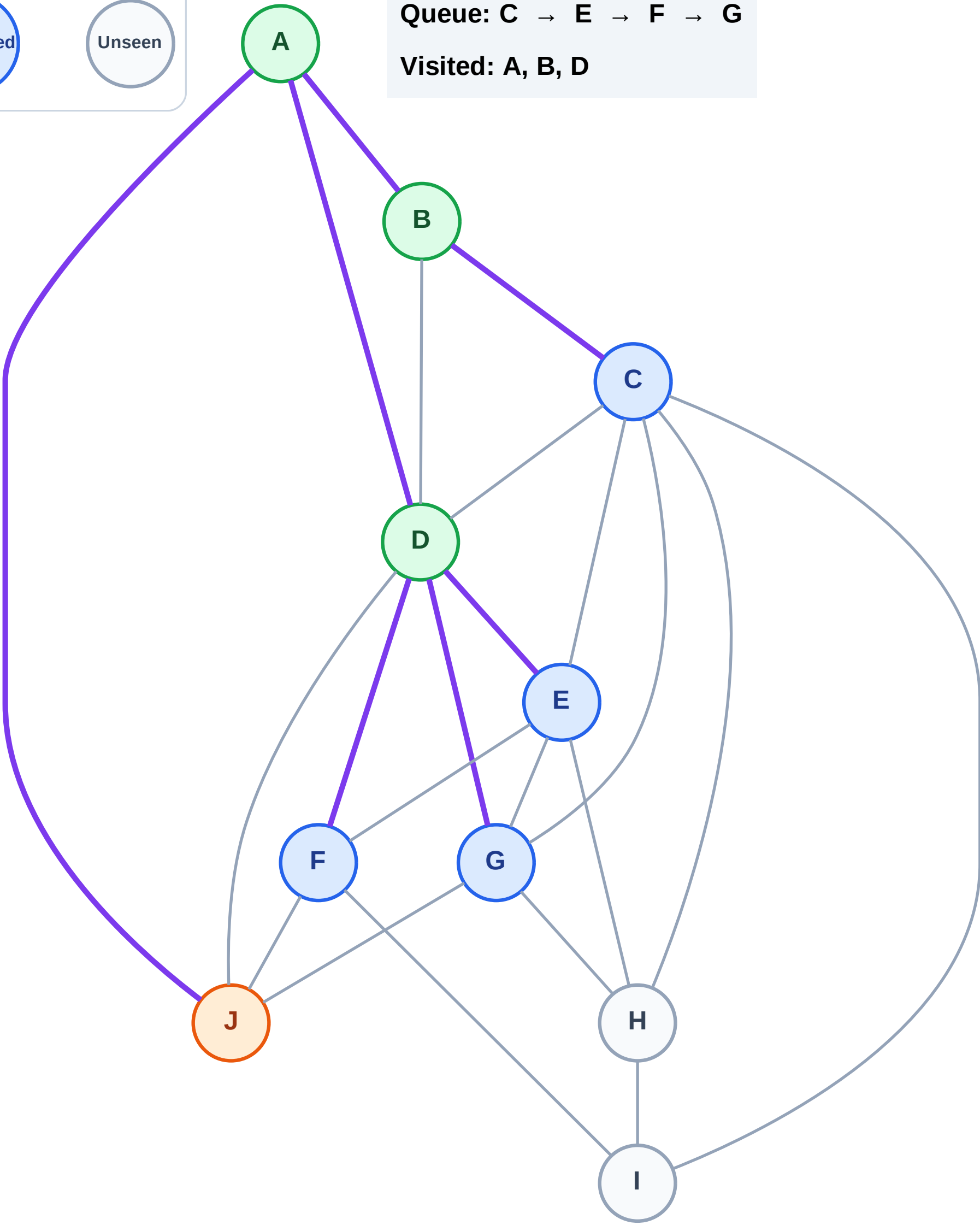
Processing

Discovered

Unseen

Queue: C → E → F → G

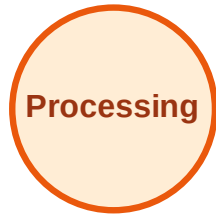
Visited: A, B, D



BFS Traversal

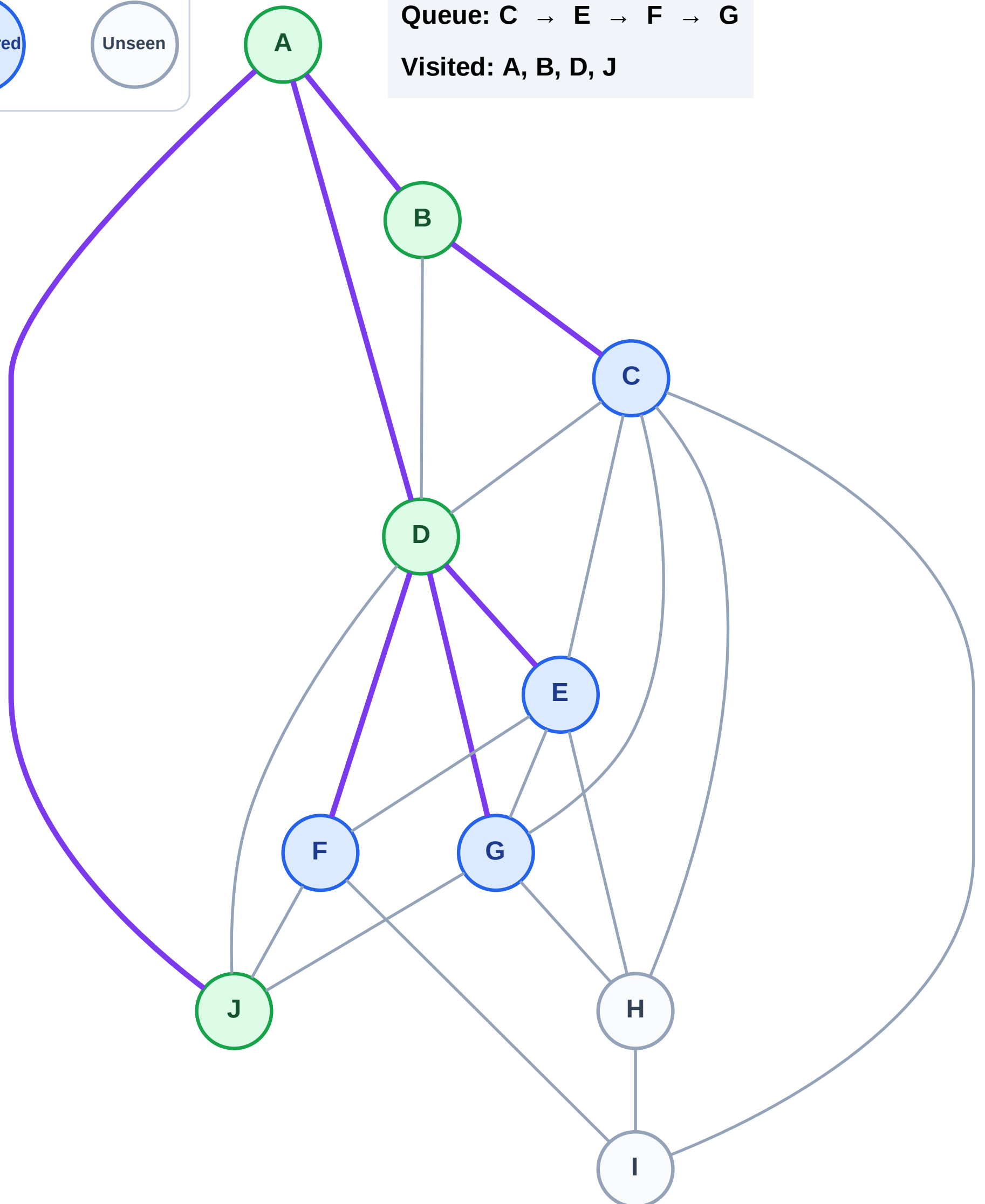
Step 9: No new neighbors from J

Legend



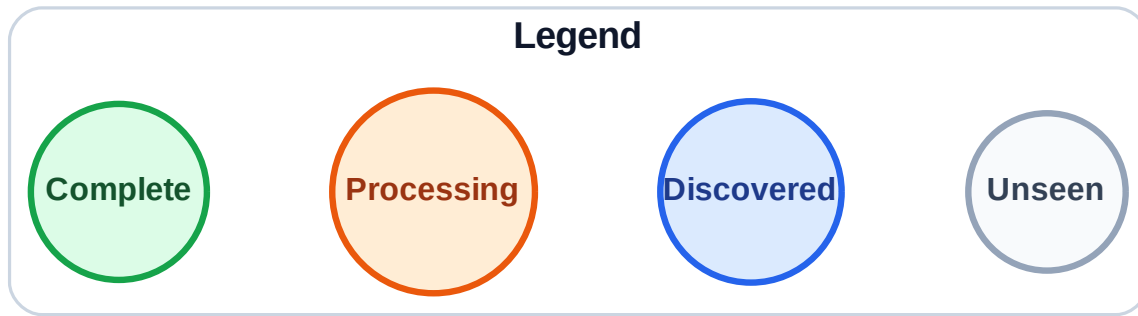
Queue: C → E → F → G

Visited: A, B, D, J



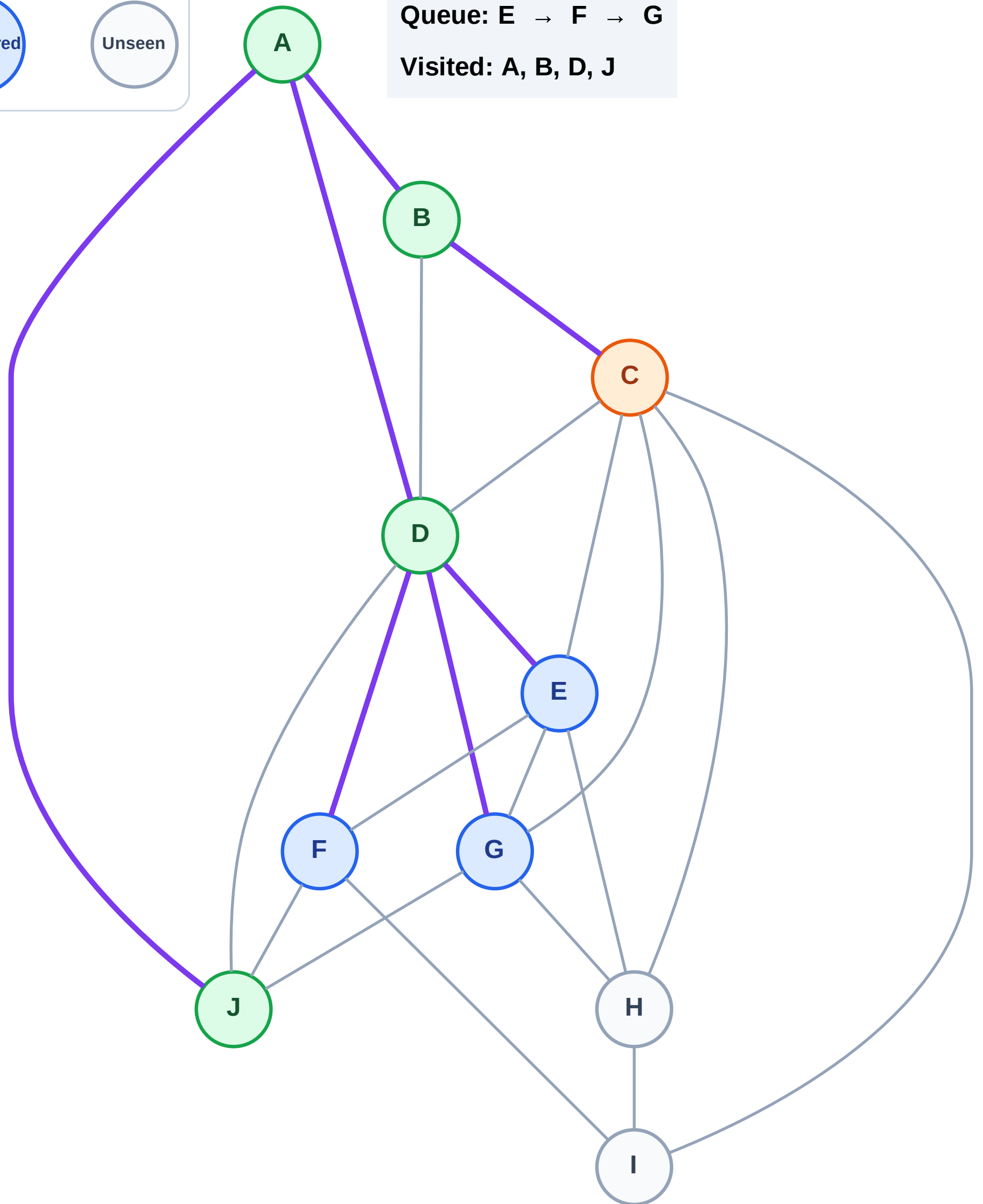
Step 10: Process node C

100%



Queue: E → F → G

Visited: A, B, D, J



BFS Traversal

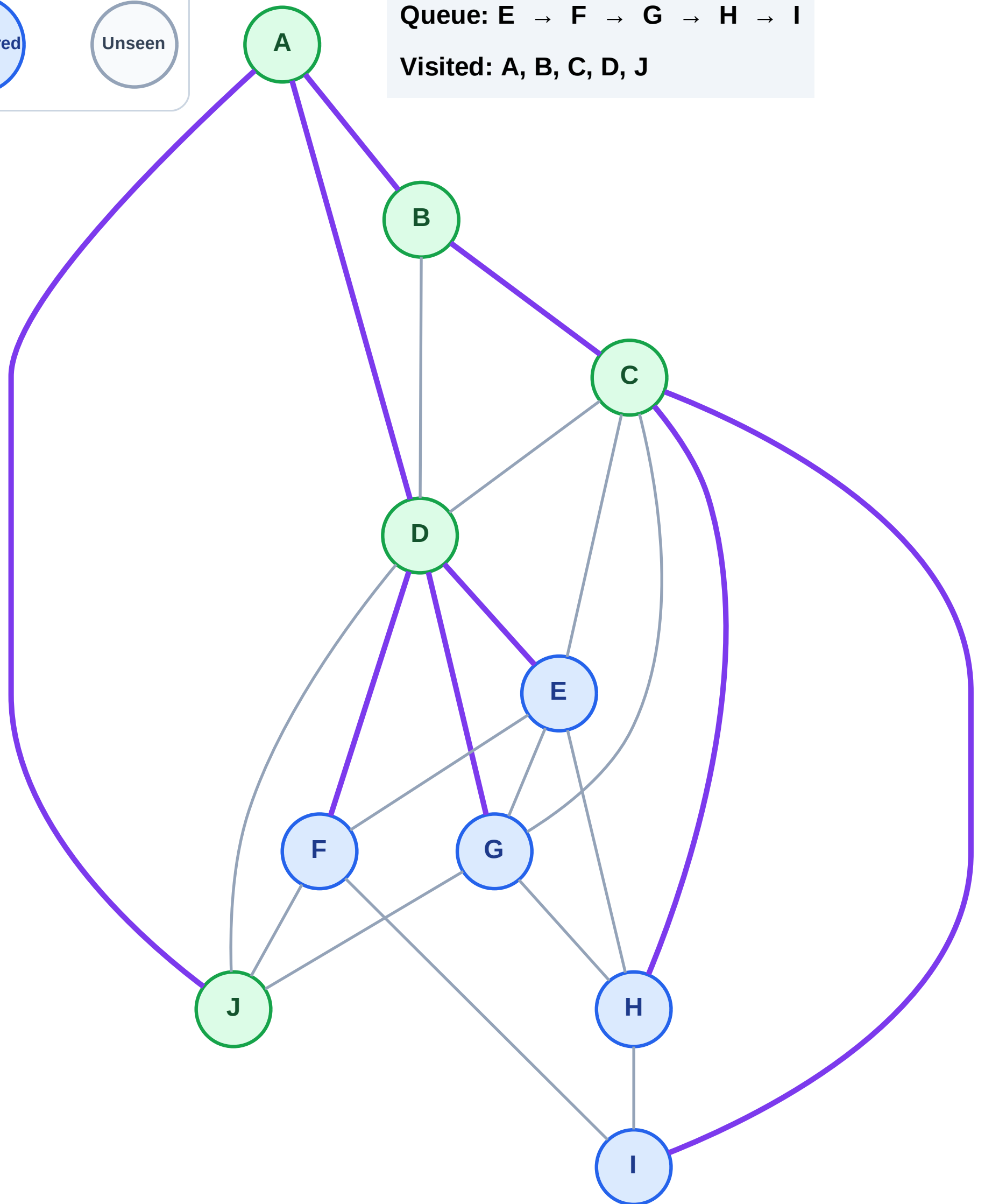
Step 11: Discover H, I from C

Legend



Queue: E → F → G → H → I

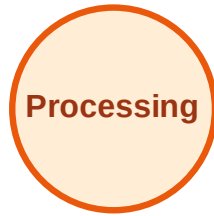
Visited: A, B, C, D, J



BFS Traversal

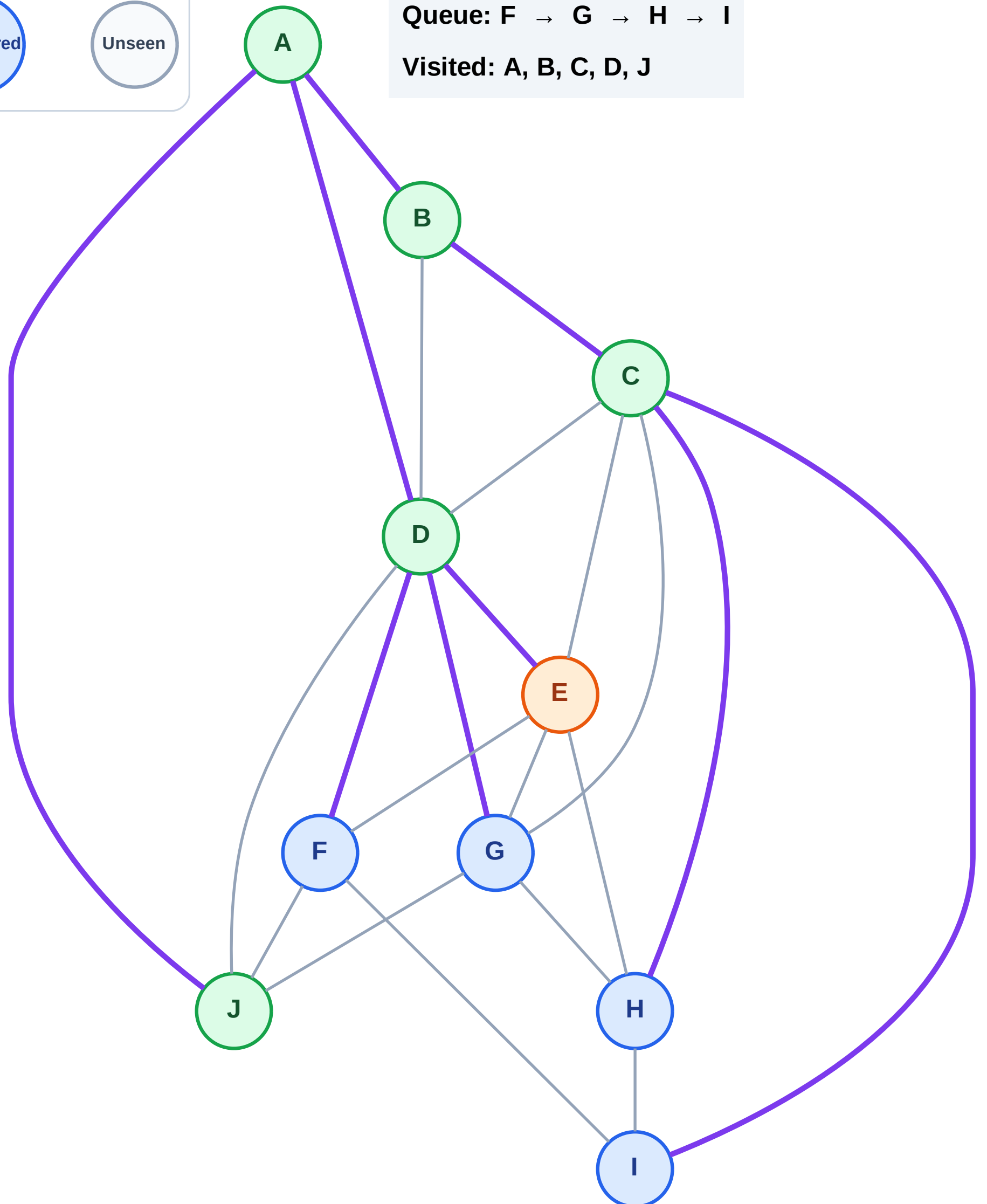
Step 12: Process node E

Legend



Queue: F → G → H → I

Visited: A, B, C, D, J



BFS Traversal

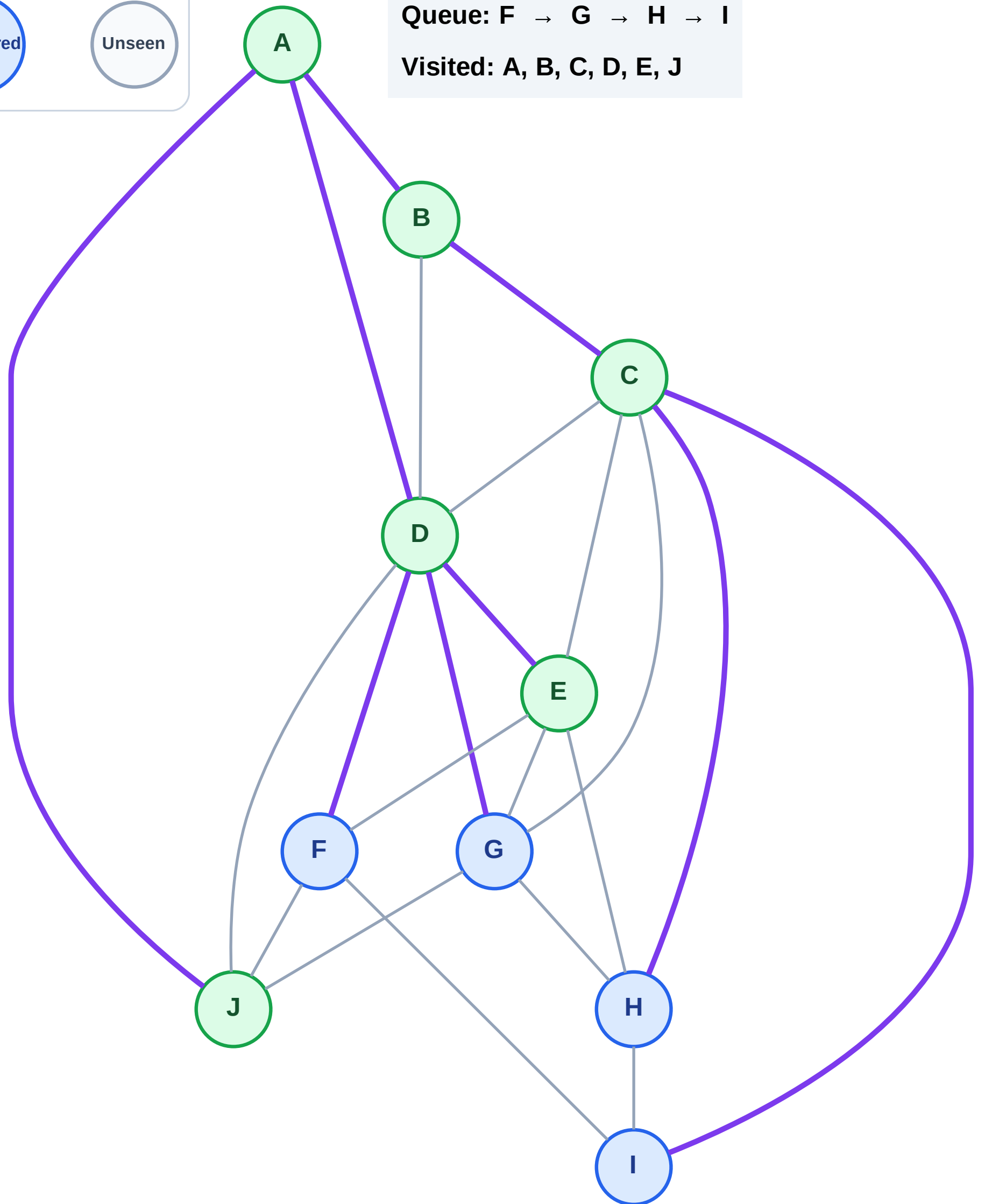
Step 13: No new neighbors from E

Legend



Queue: F → G → H → I

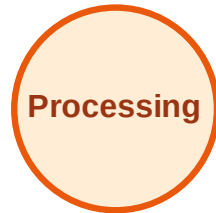
Visited: A, B, C, D, E, J



BFS Traversal

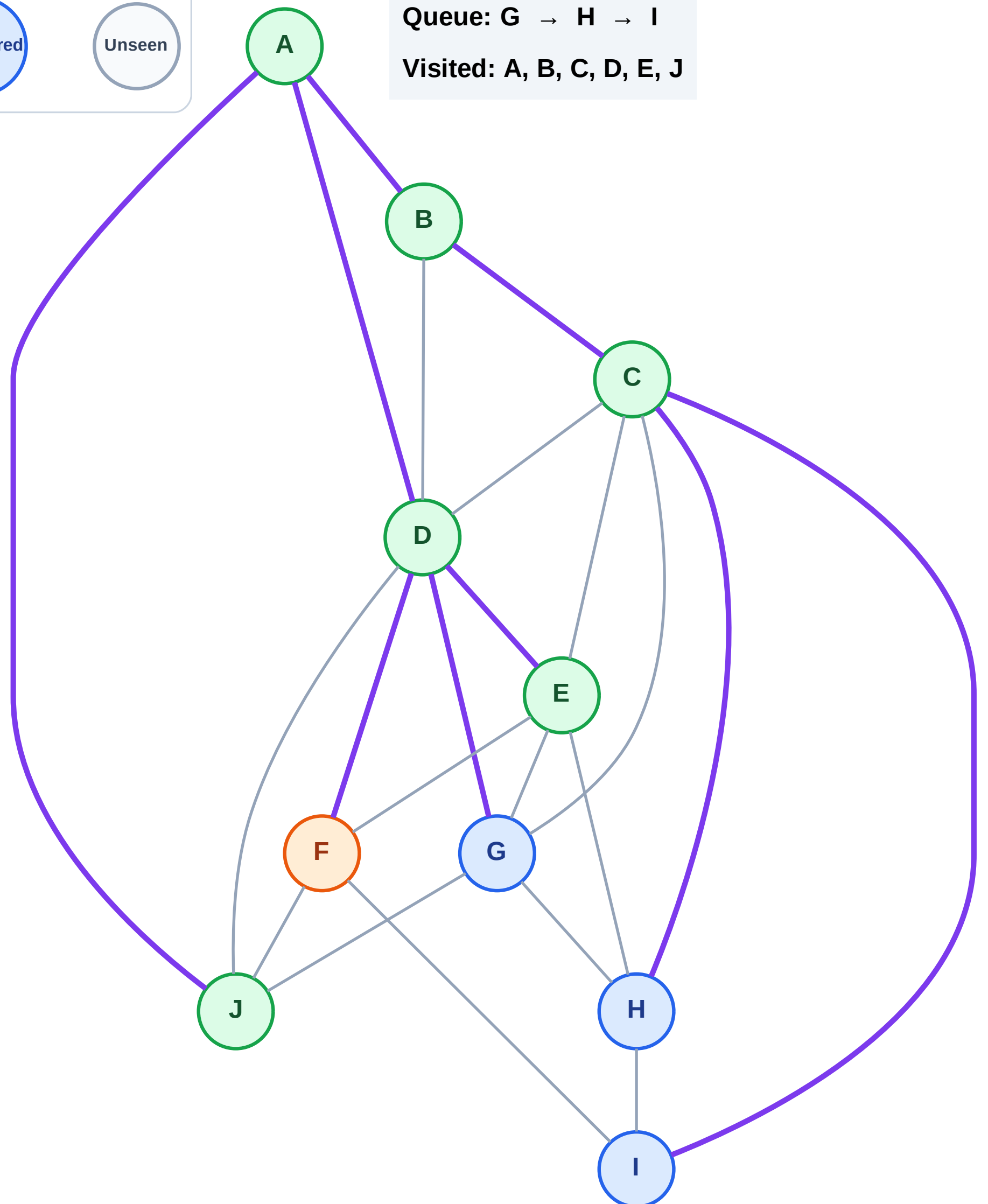
Step 14: Process node F

Legend



Queue: G → H → I

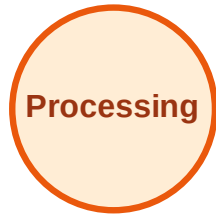
Visited: A, B, C, D, E, J



BFS Traversal

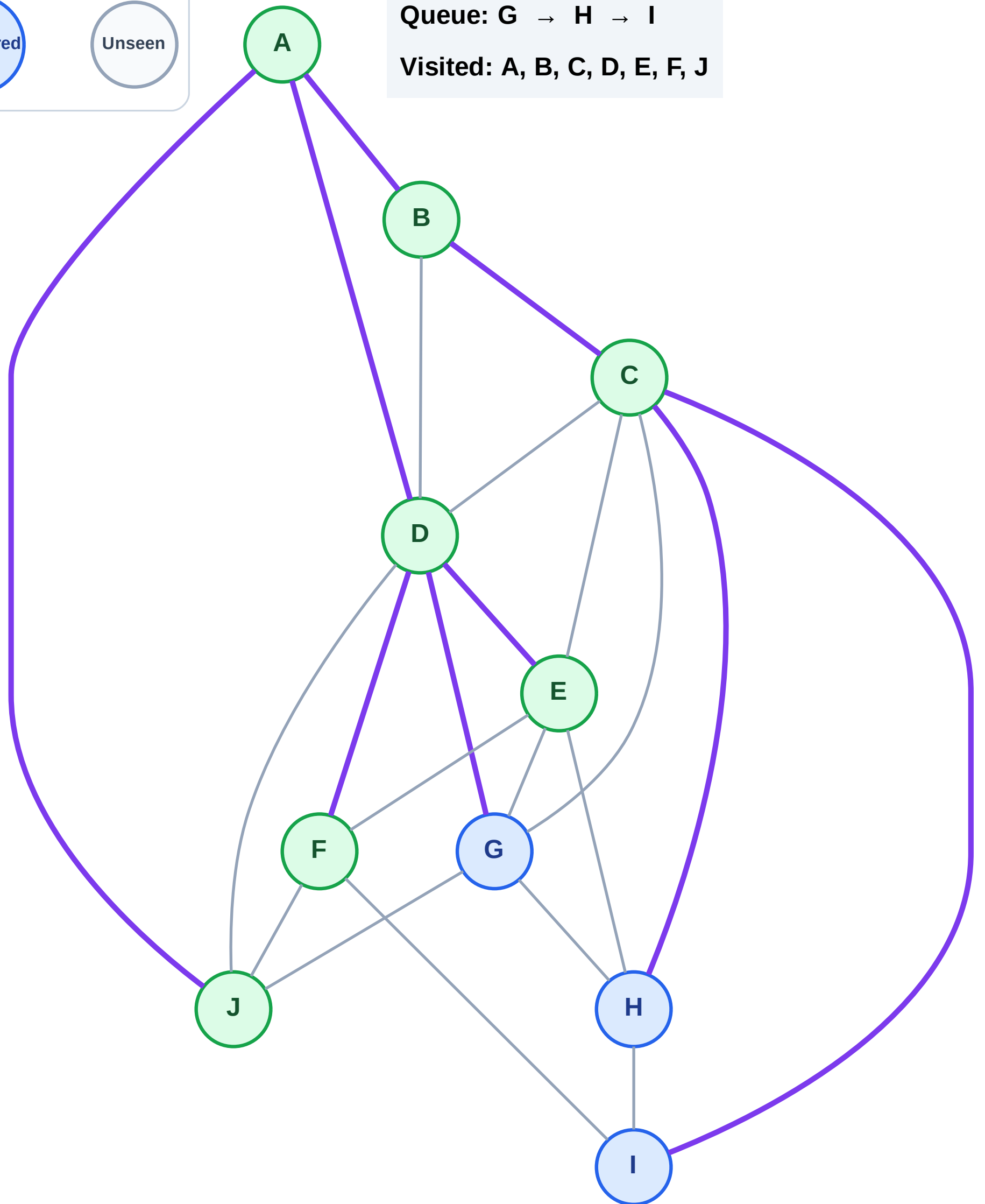
Step 15: No new neighbors from F

Legend



Queue: G → H → I

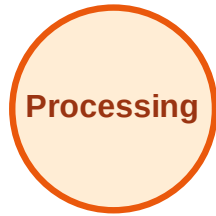
Visited: A, B, C, D, E, F, J



BFS Traversal

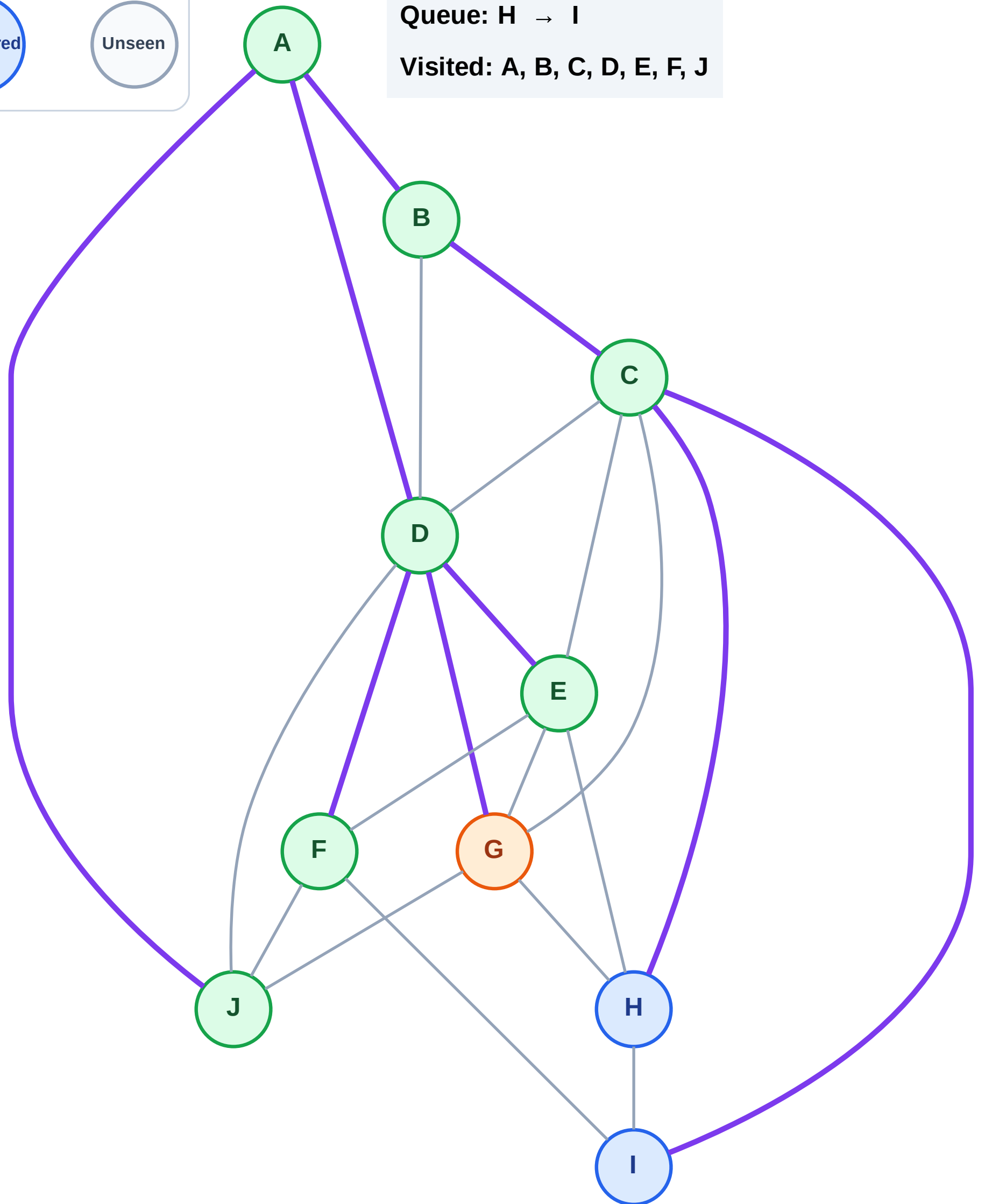
Step 16: Process node G

Legend



Queue: H → I

Visited: A, B, C, D, E, F, J



BFS Traversal

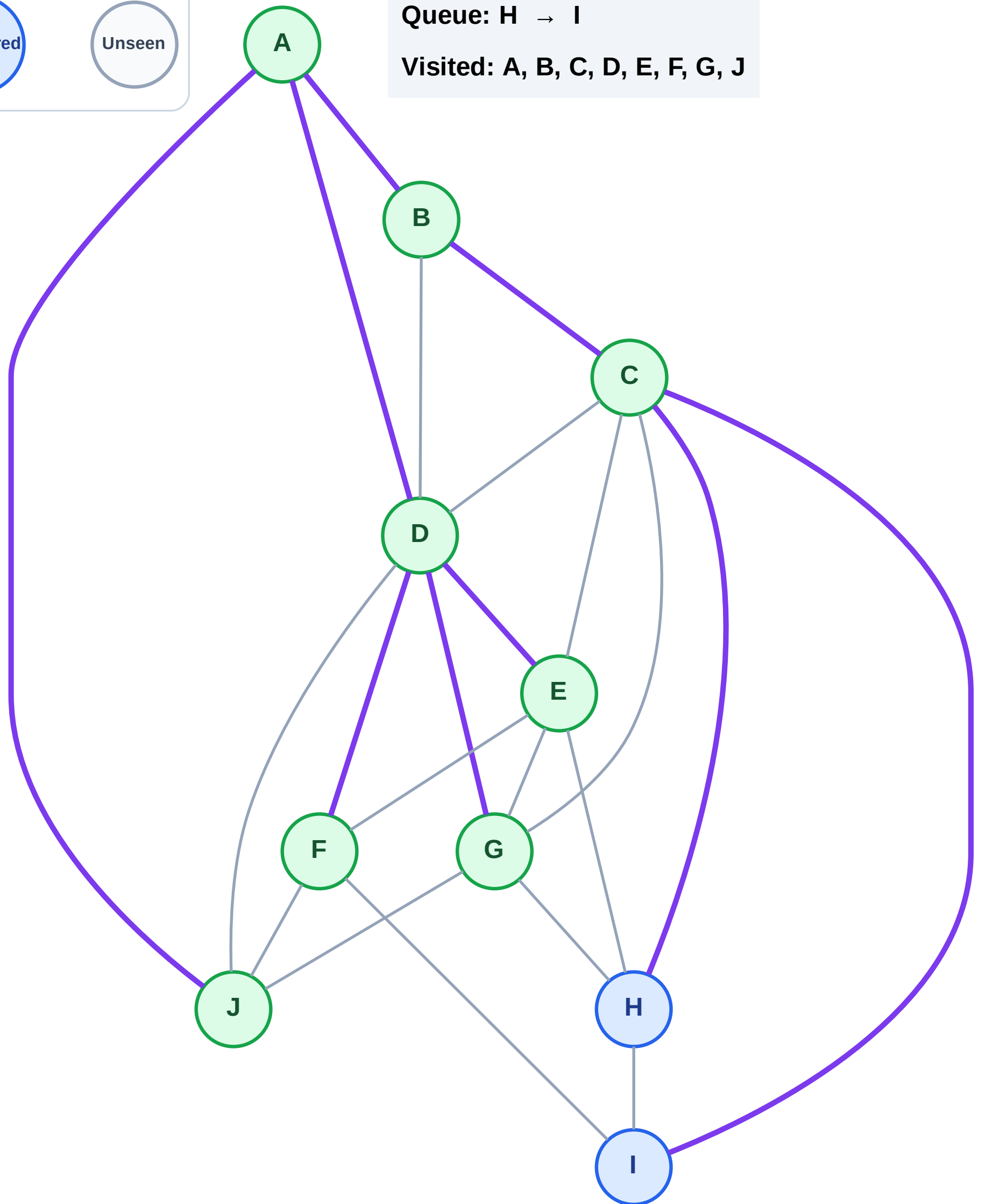
Step 17: No new neighbors from G

Legend



Queue: H → I

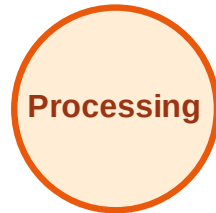
Visited: A, B, C, D, E, F, G, J



BFS Traversal

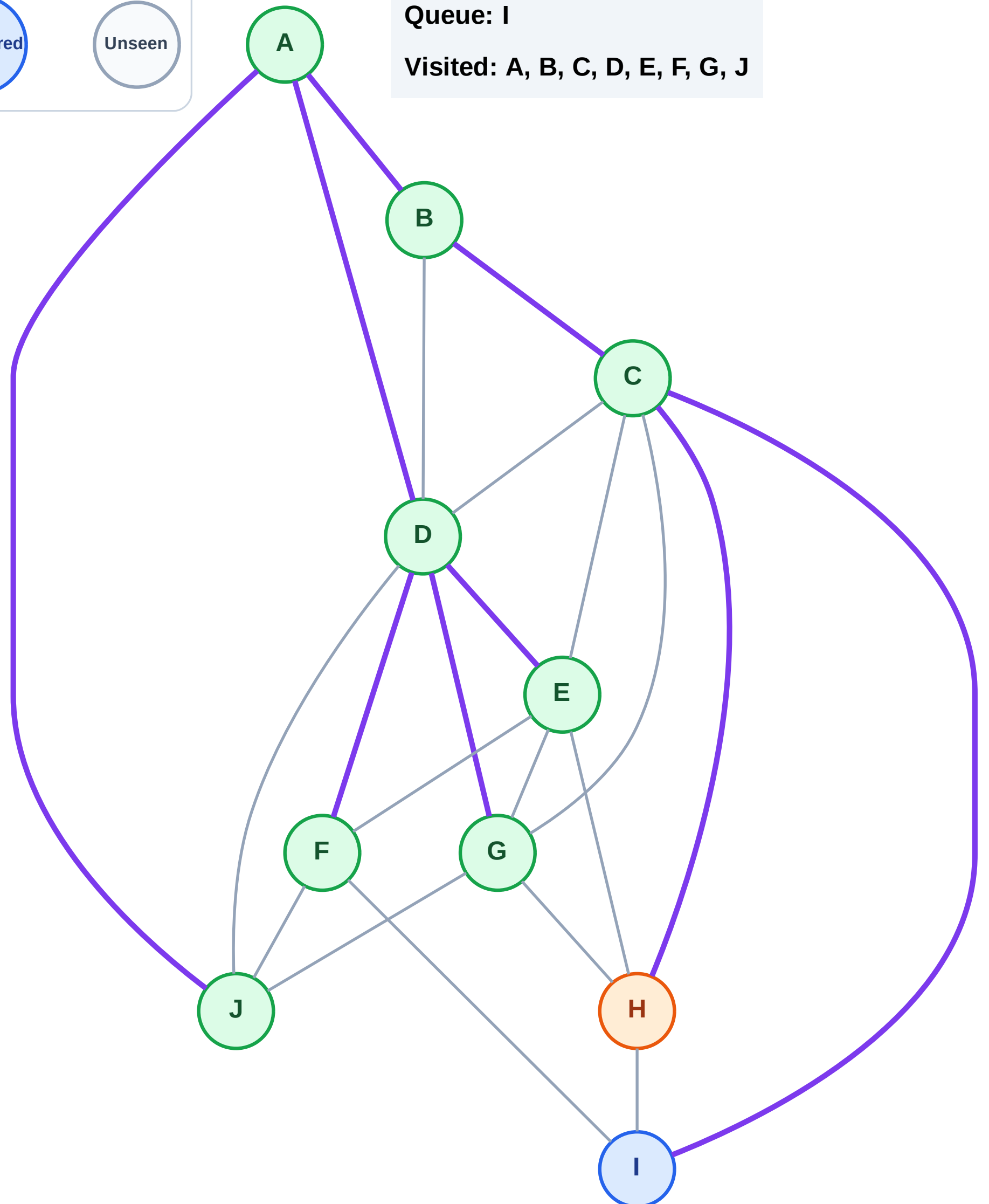
Step 18: Process node H

Legend



Queue: I

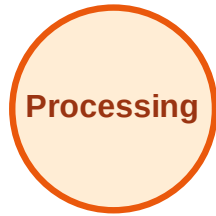
Visited: A, B, C, D, E, F, G, J



BFS Traversal

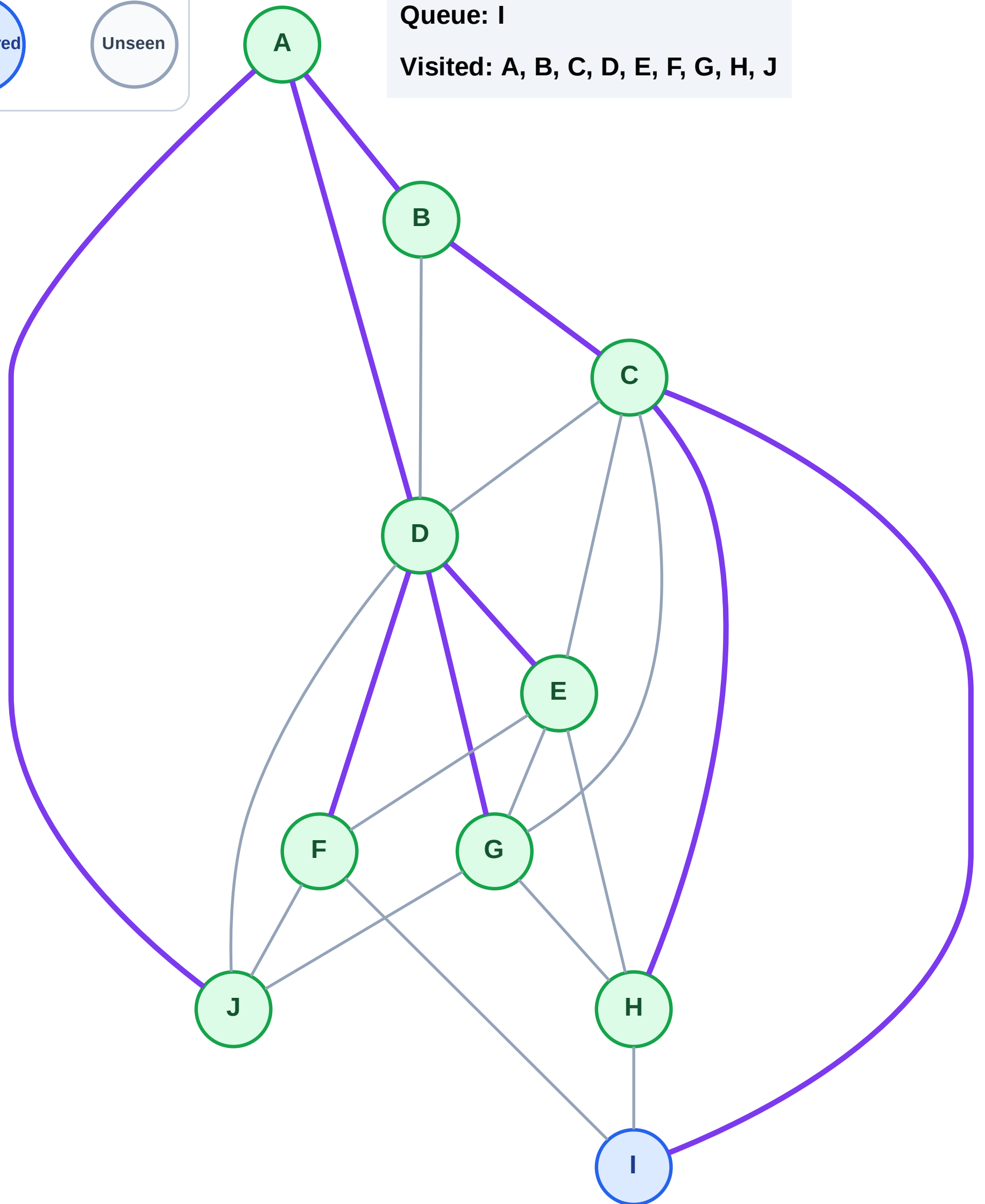
Step 19: No new neighbors from H

Legend



Queue: I

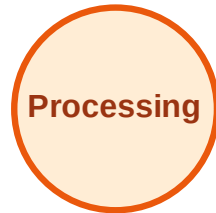
Visited: A, B, C, D, E, F, G, H, J



BFS Traversal

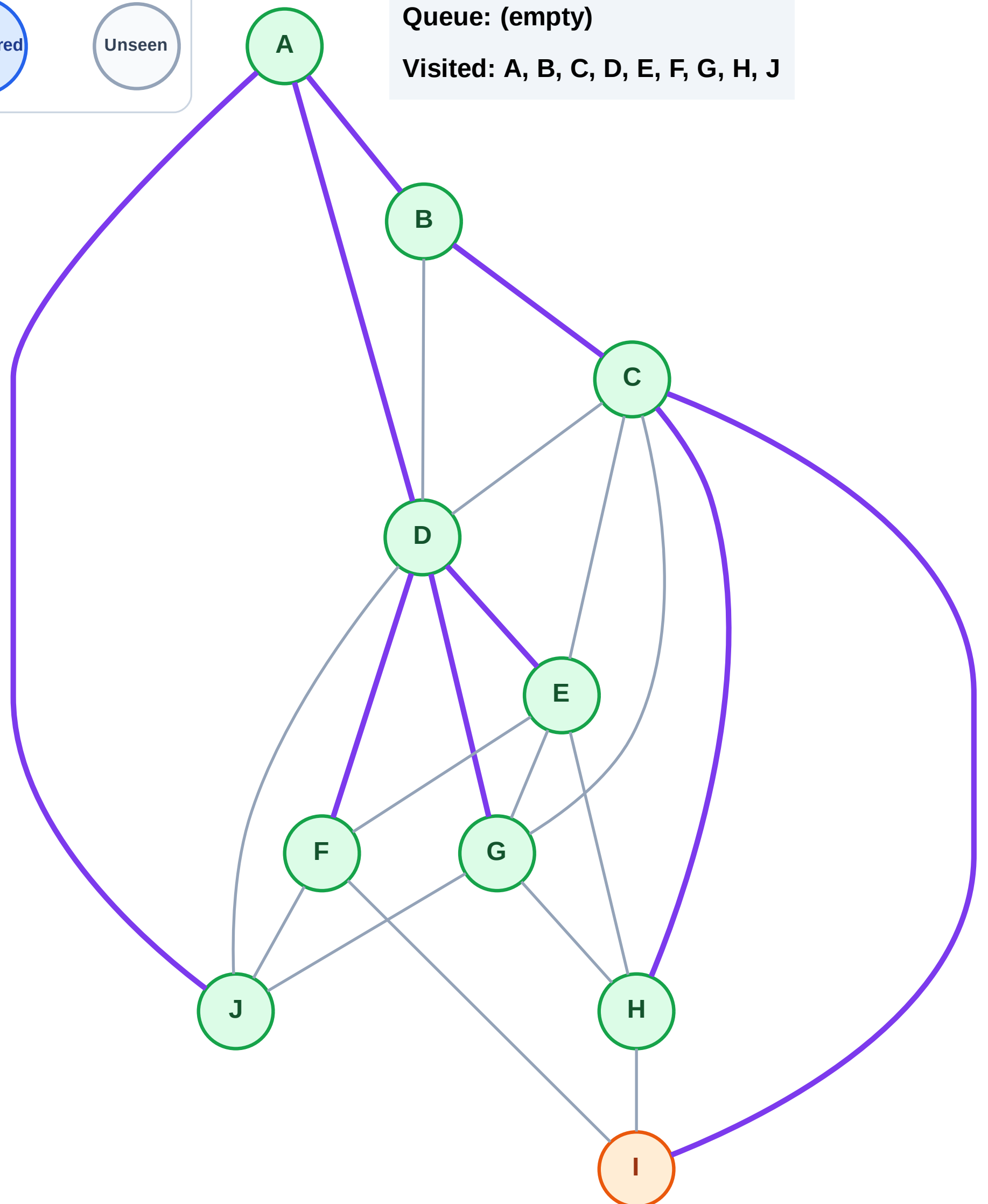
Step 20: Process node I

Legend



Queue: (empty)

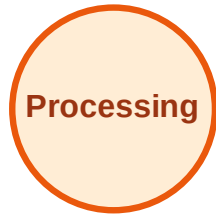
Visited: A, B, C, D, E, F, G, H, J



BFS Traversal

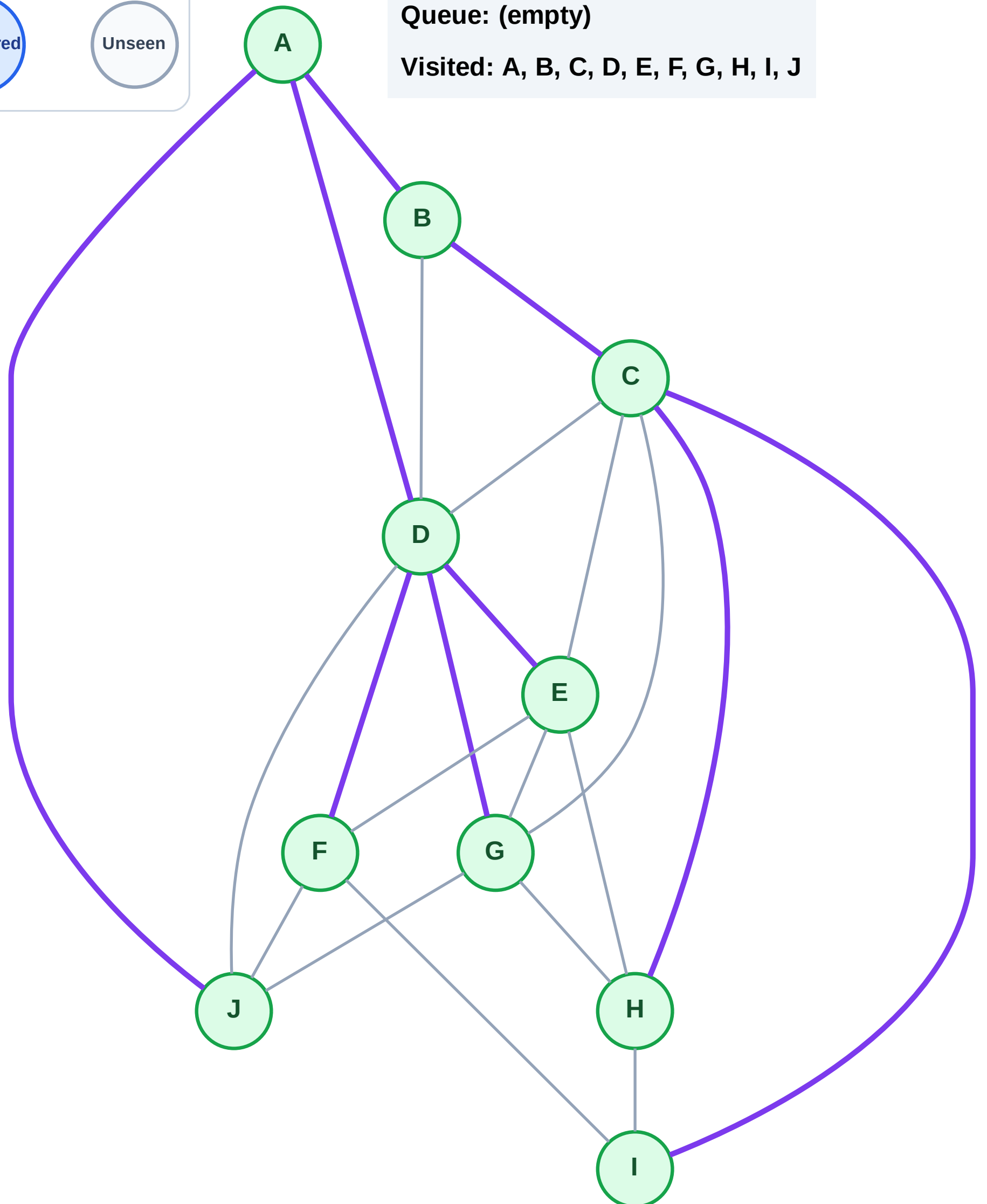
Step 21: No new neighbors from I

Legend



Queue: (empty)

Visited: A, B, C, D, E, F, G, H, I, J



DFS Traversal

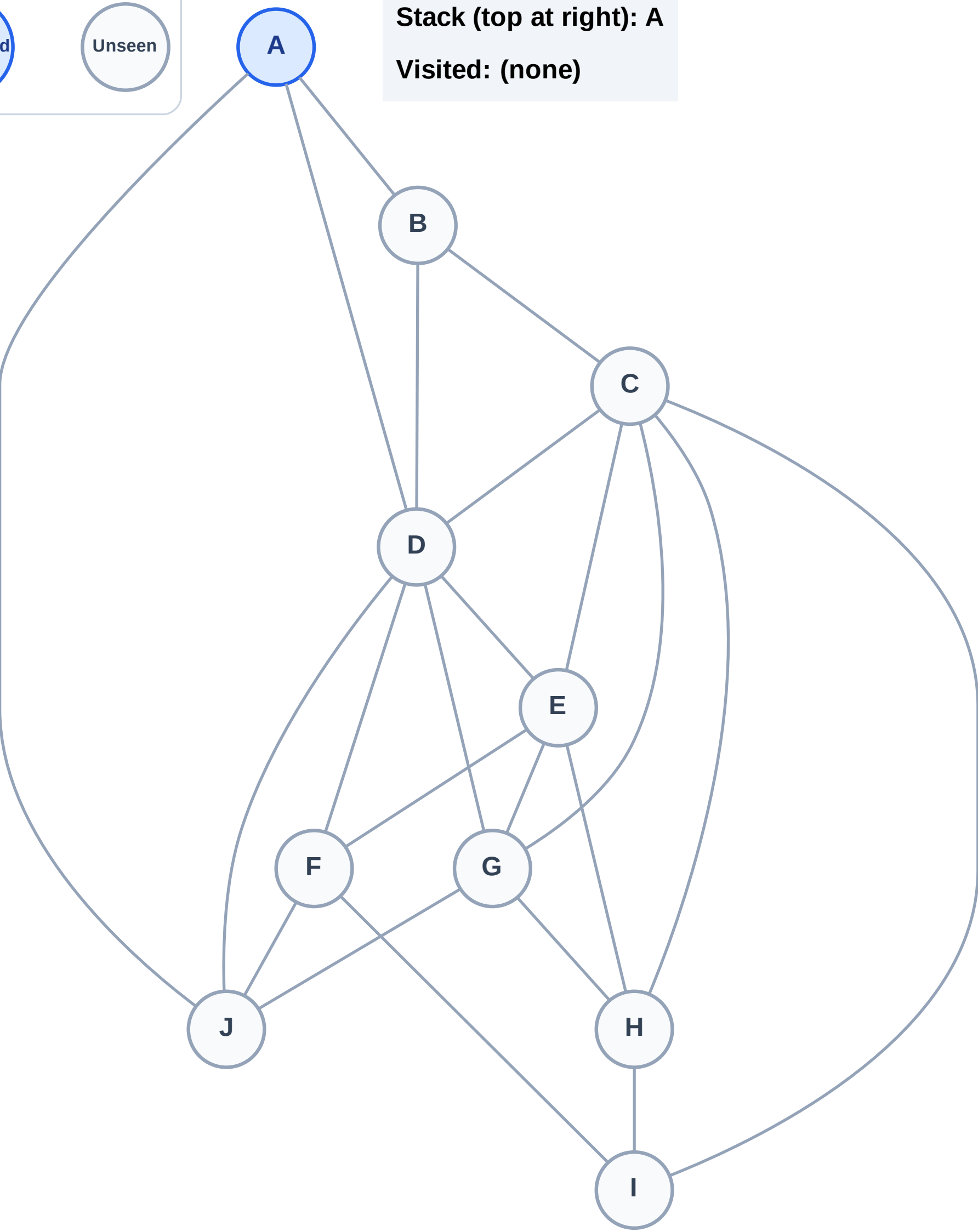
Step 1: Start at A

Legend



Stack (top at right): A

Visited: (none)



DFS Traversal

Step 2: Process node A

Legend

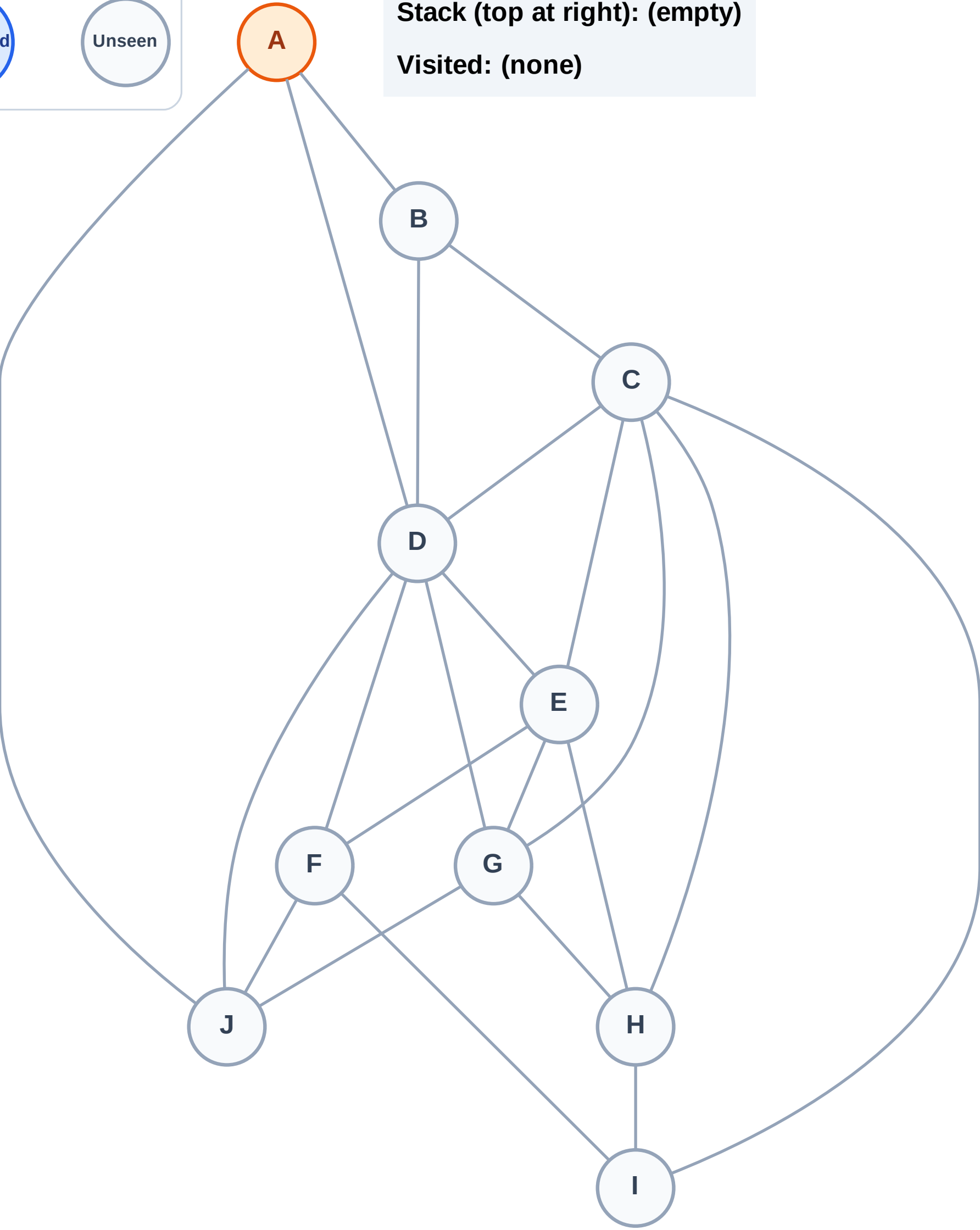
Complete

Processing

Discovered

Unseen

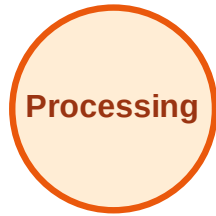
Stack (top at right): (empty)
Visited: (none)



DFS Traversal

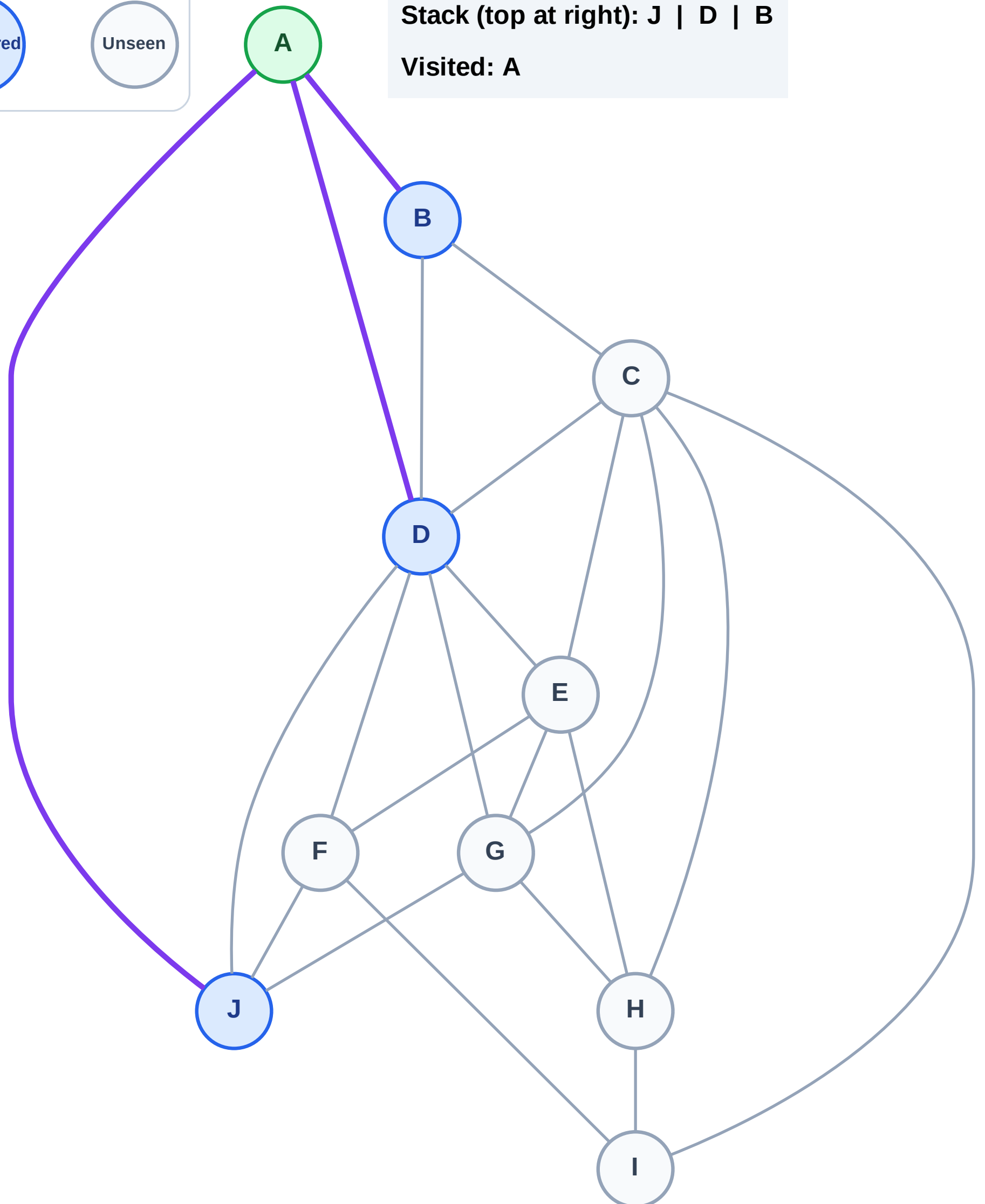
Step 3: Discover J, D, B from A

Legend



Stack (top at right): J | D | B

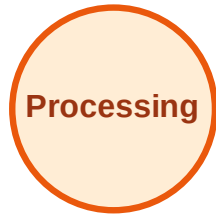
Visited: A



DFS Traversal

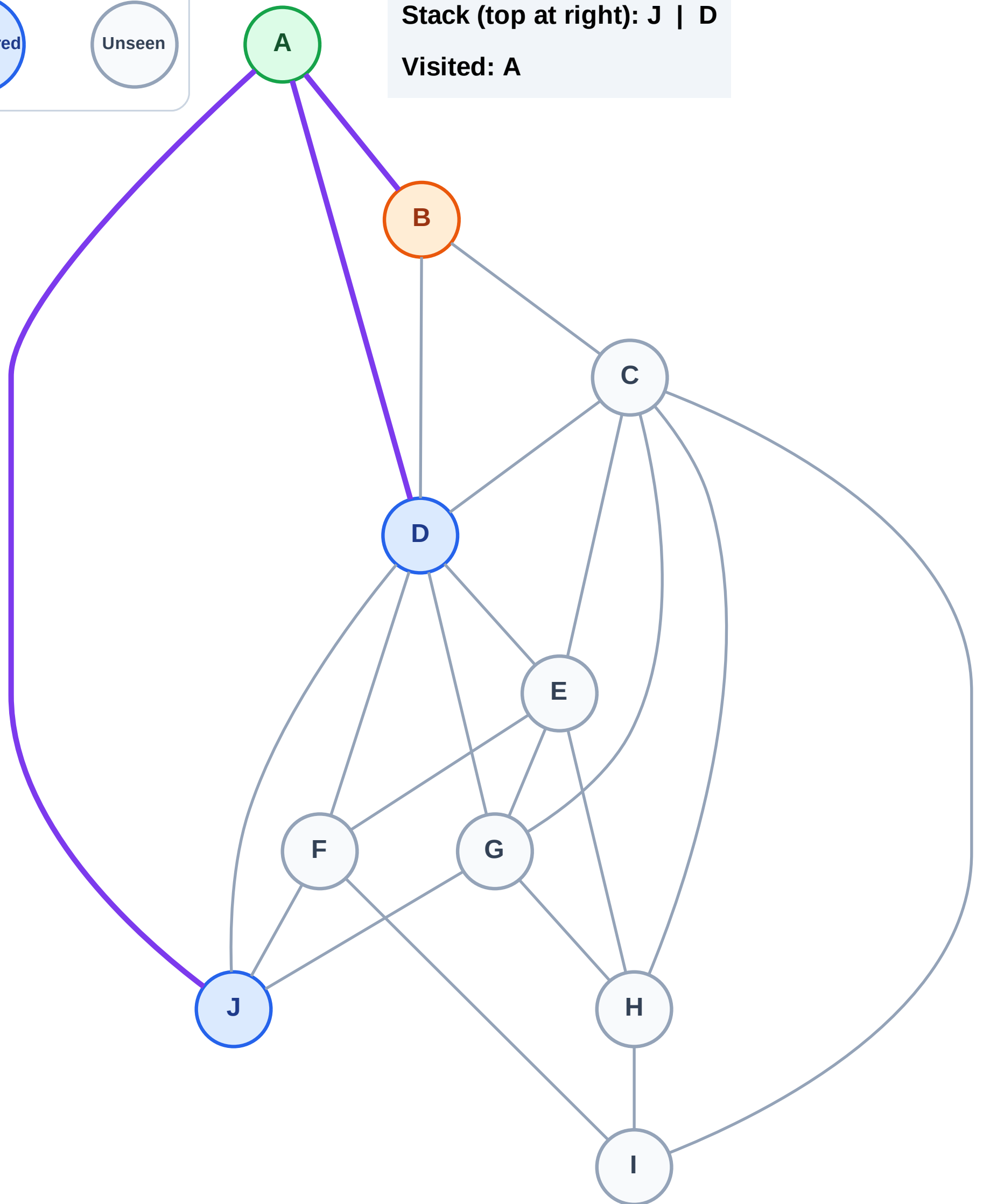
Step 4: Process node B

Legend



Stack (top at right): J | D

Visited: A



DFS Traversal

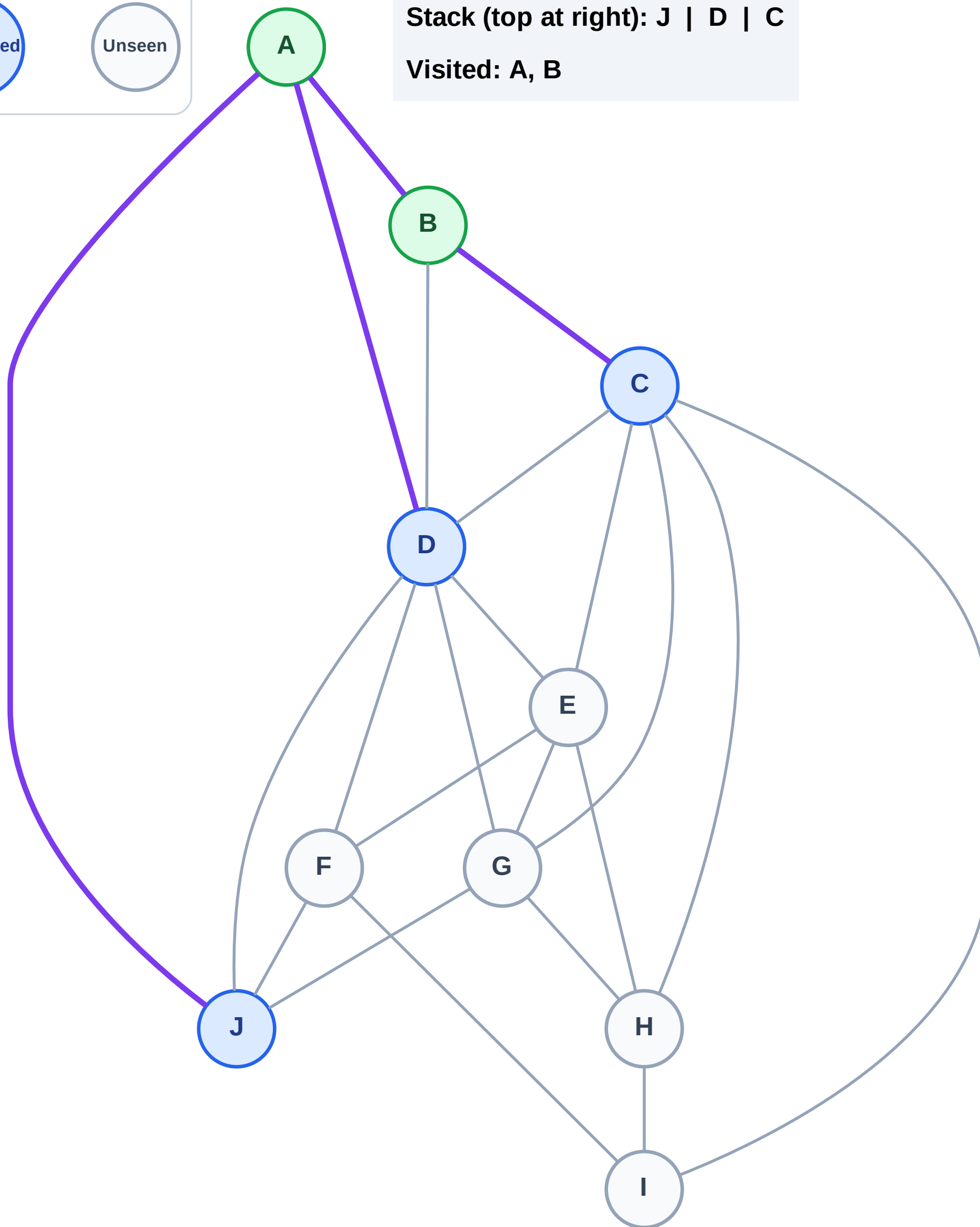
Step 5: Discover C from B

Legend



Stack (top at right): J | D | C

Visited: A, B



DFS Traversal

Step 6: Process node C

Legend

Complete

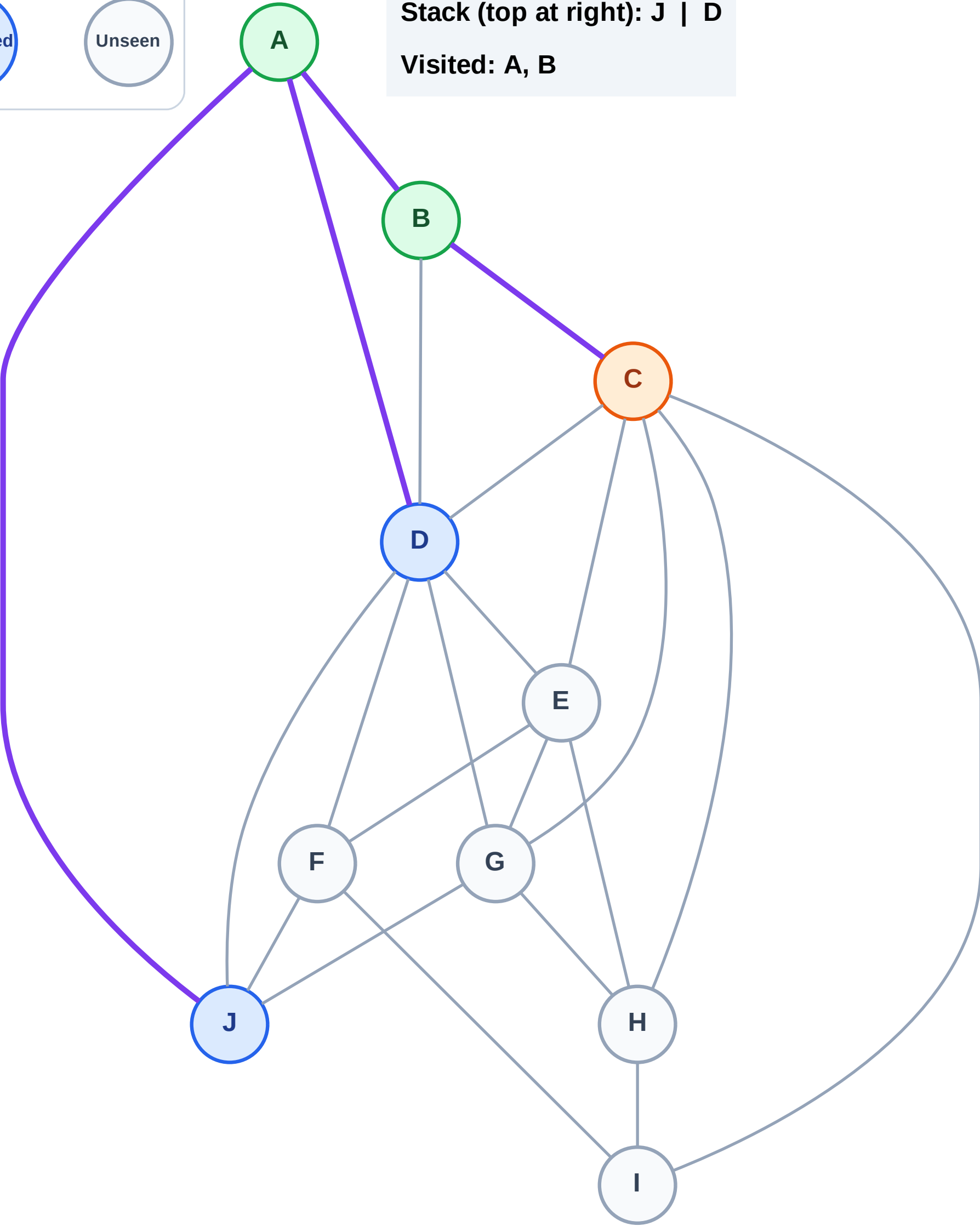
Processing

Discovered

Unseen

Stack (top at right): J | D

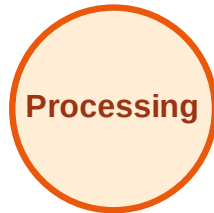
Visited: A, B



DFS Traversal

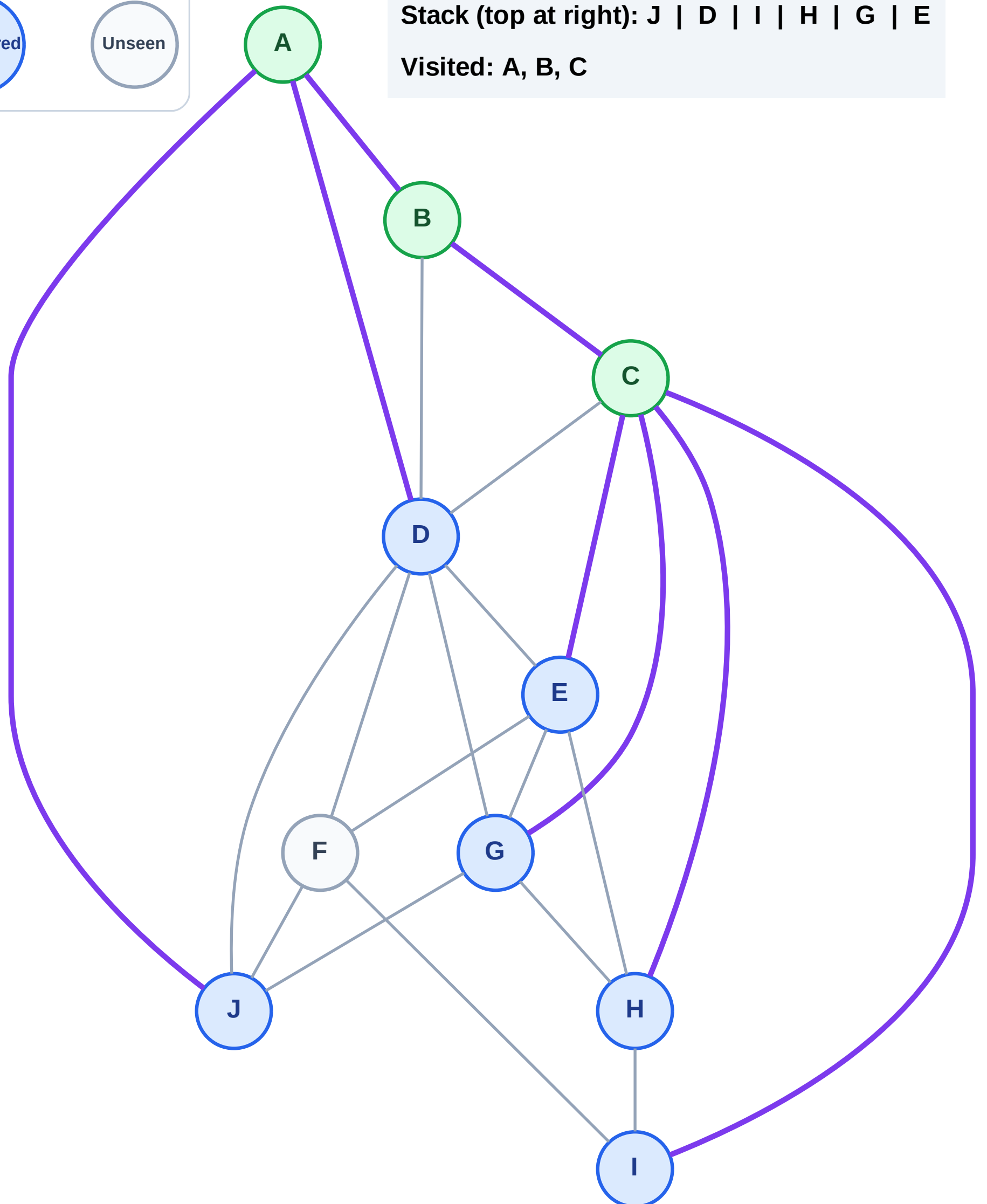
Step 7: Discover I, H, G, E from C

Legend

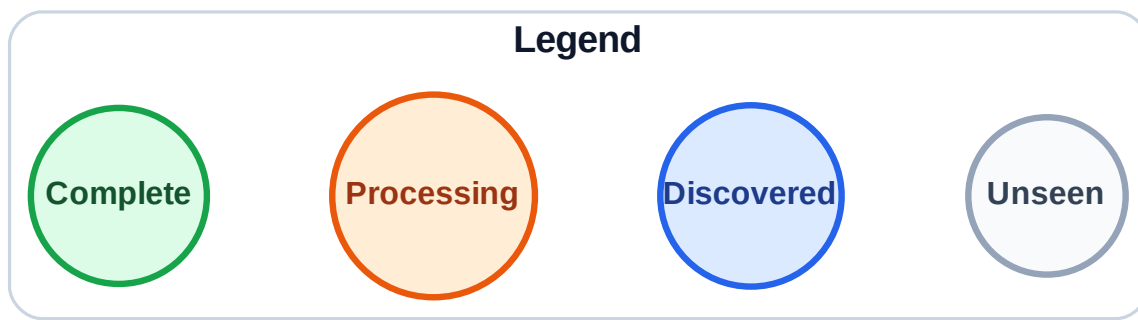


Stack (top at right): J | D | I | H | G | E

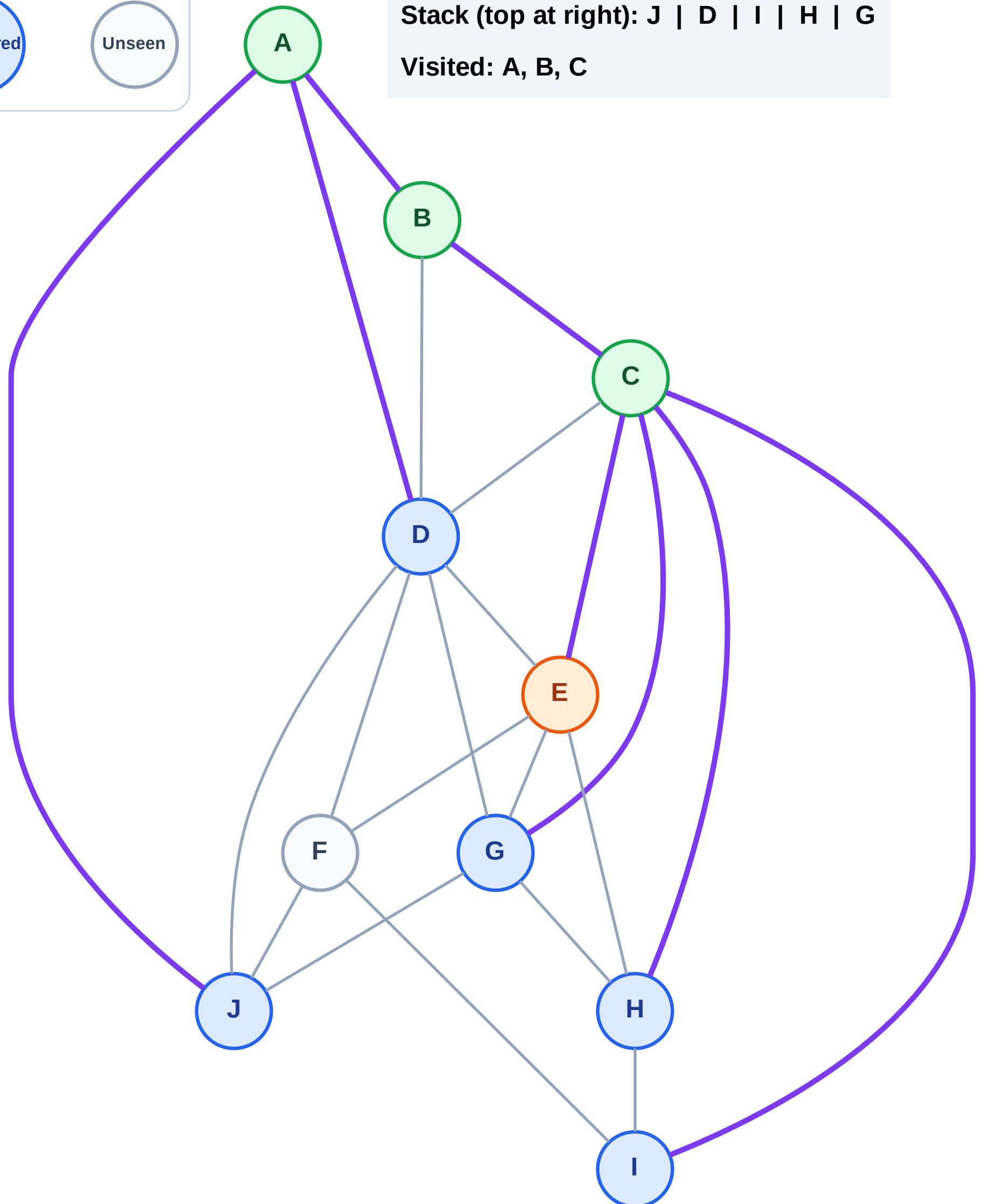
Visited: A, B, C



Step 8: Process node E



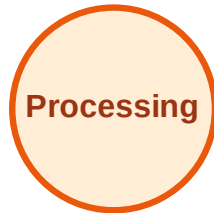
Visited: A, B, C



DFS Traversal

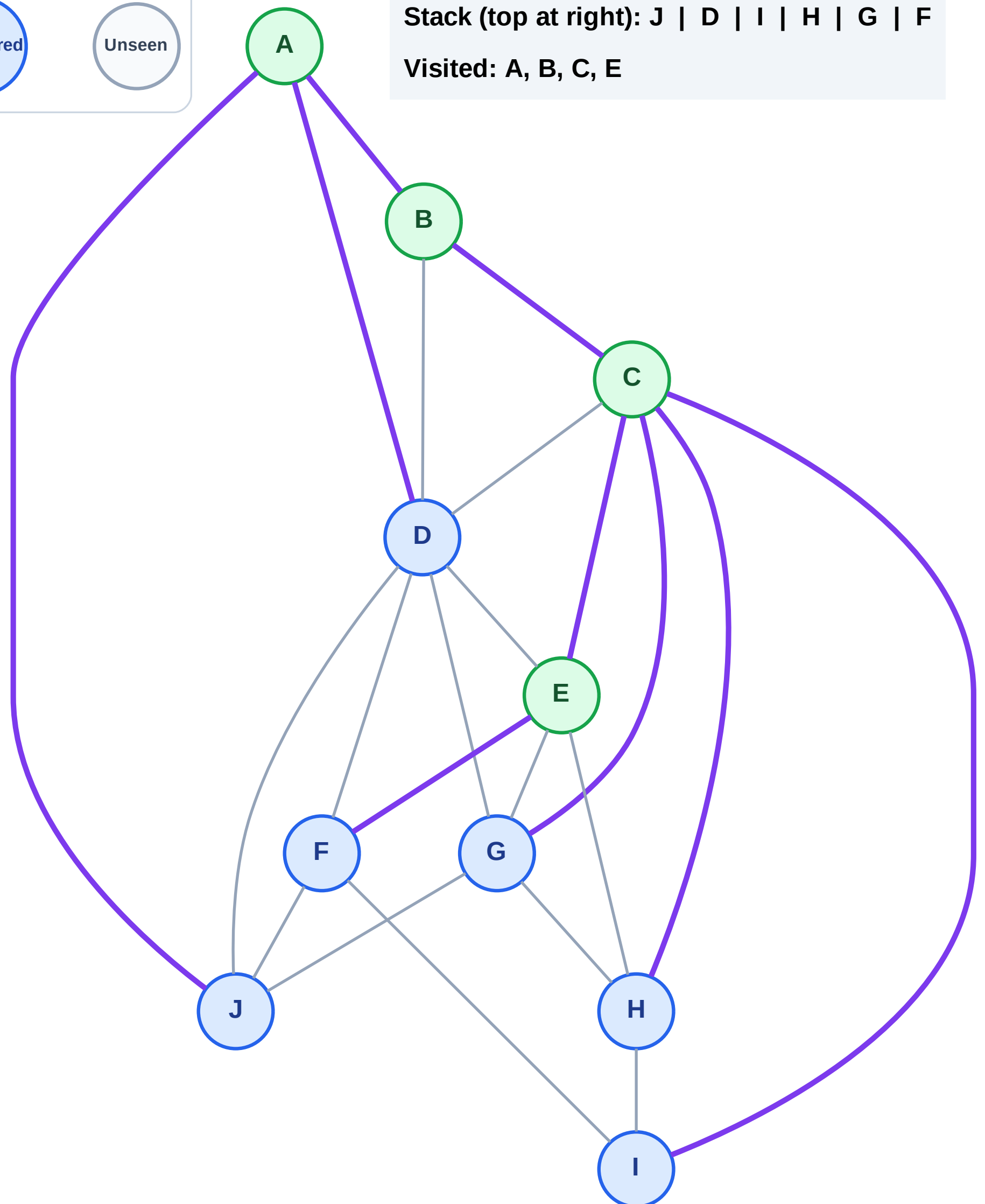
Step 9: Discover F from E

Legend



Stack (top at right): J | D | I | H | G | F

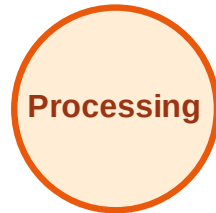
Visited: A, B, C, E



DFS Traversal

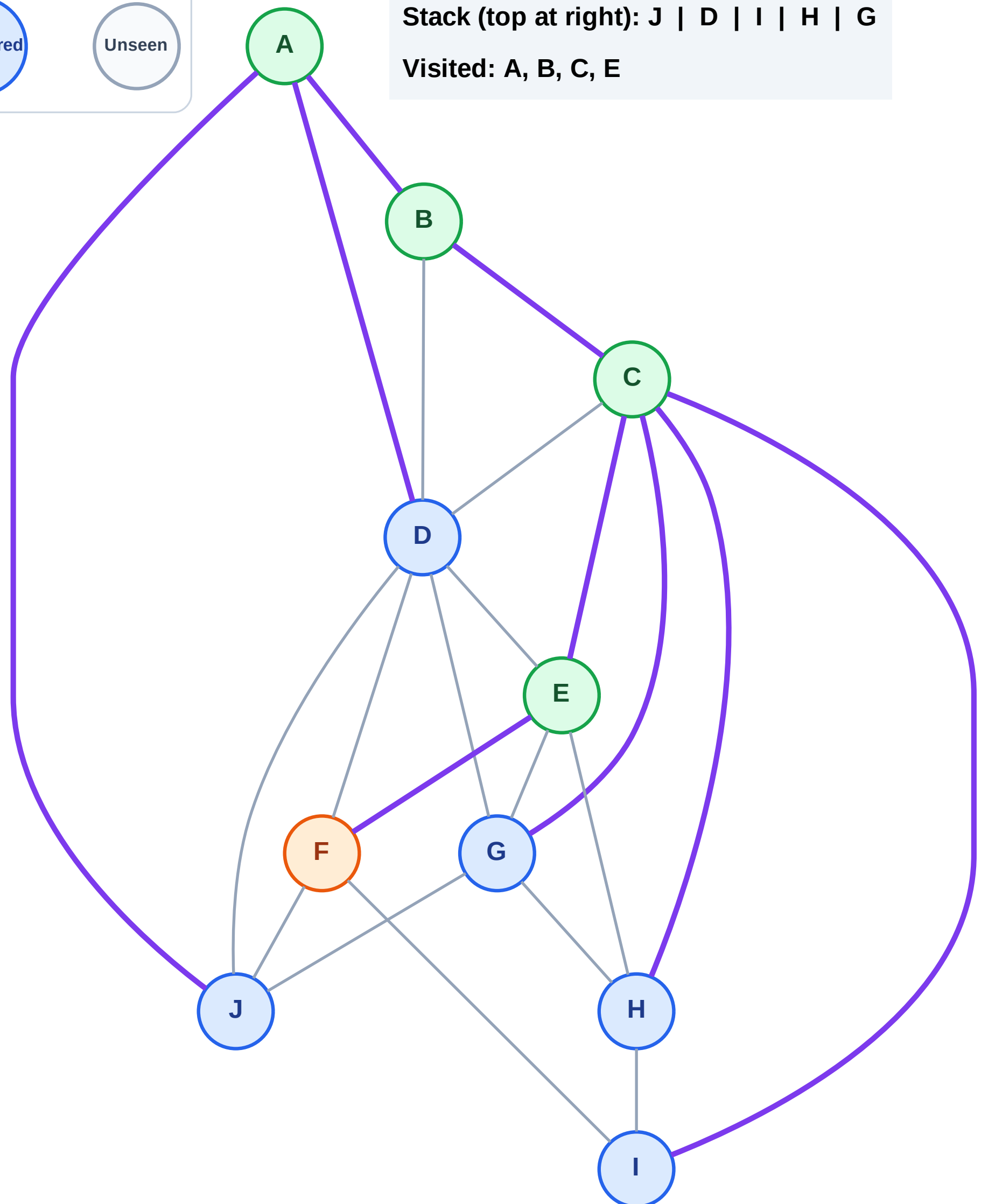
Step 10: Process node F

Legend



Stack (top at right): J | D | I | H | G

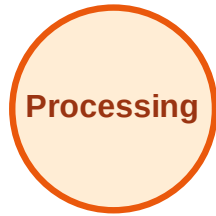
Visited: A, B, C, E



DFS Traversal

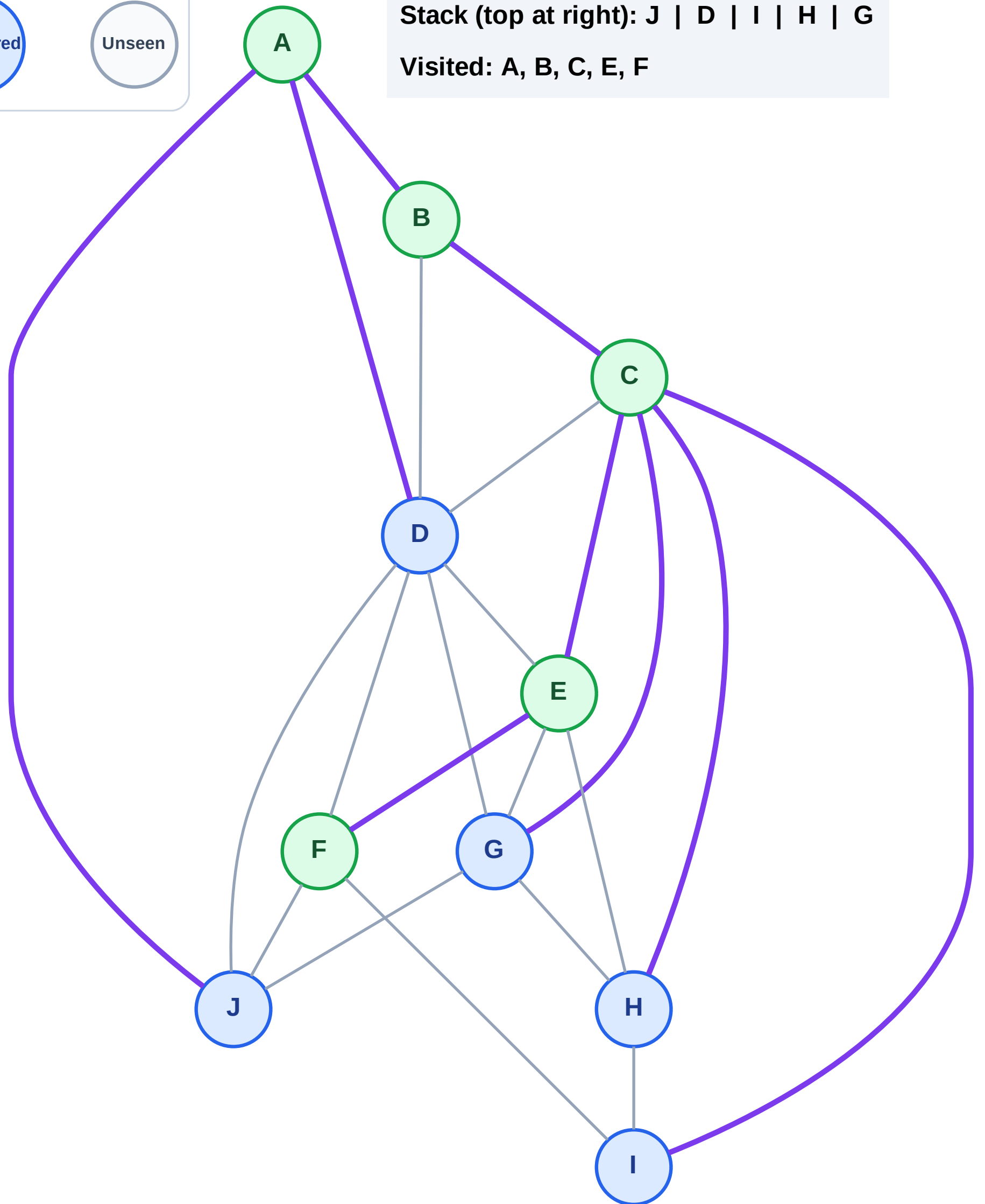
Step 11: No new neighbors from F

Legend



Stack (top at right): J | D | I | H | G

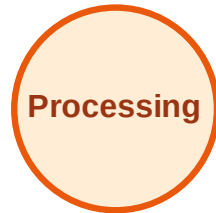
Visited: A, B, C, E, F



DFS Traversal

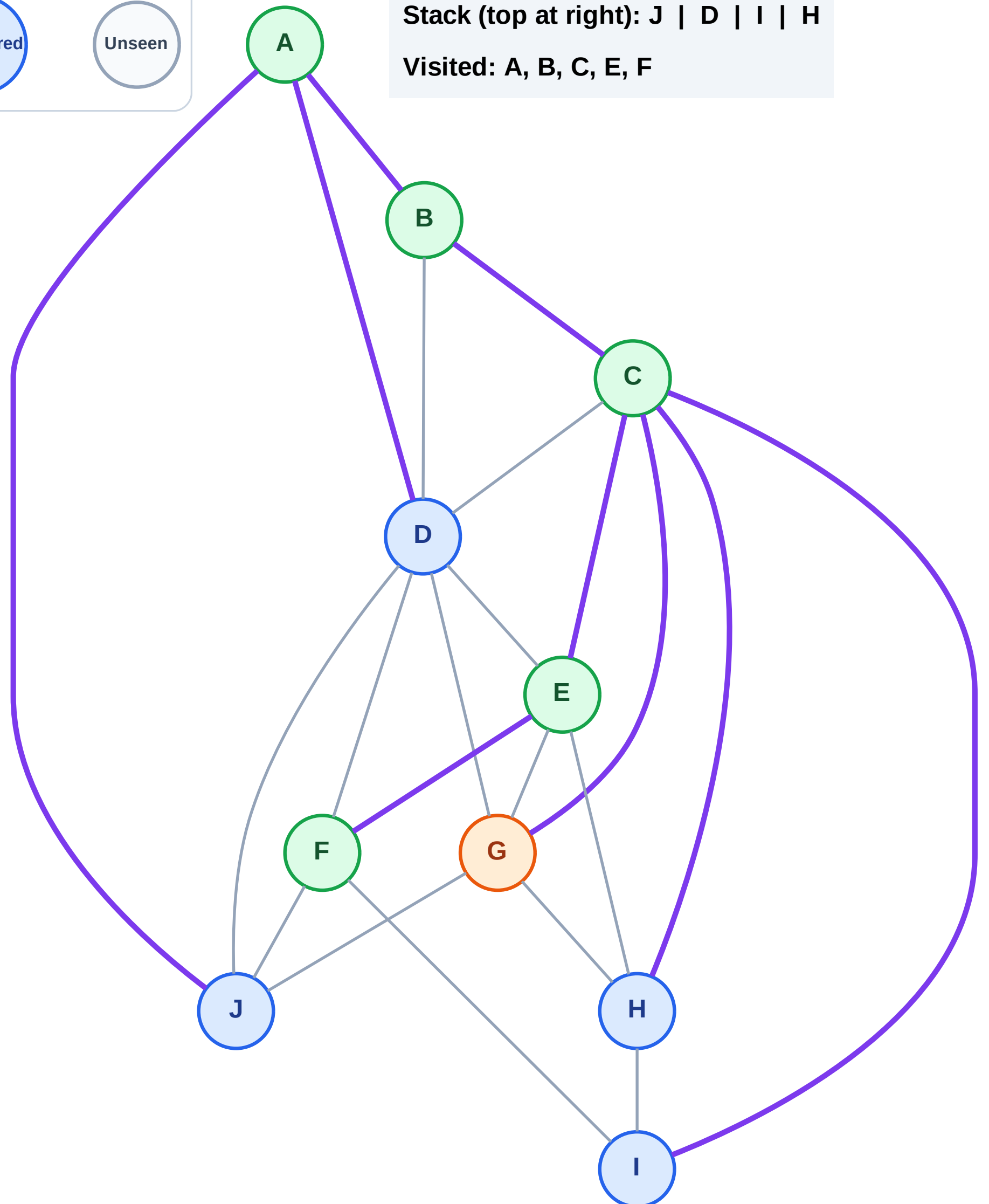
Step 12: Process node G

Legend



Stack (top at right): J | D | I | H

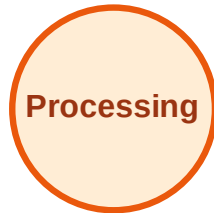
Visited: A, B, C, E, F



DFS Traversal

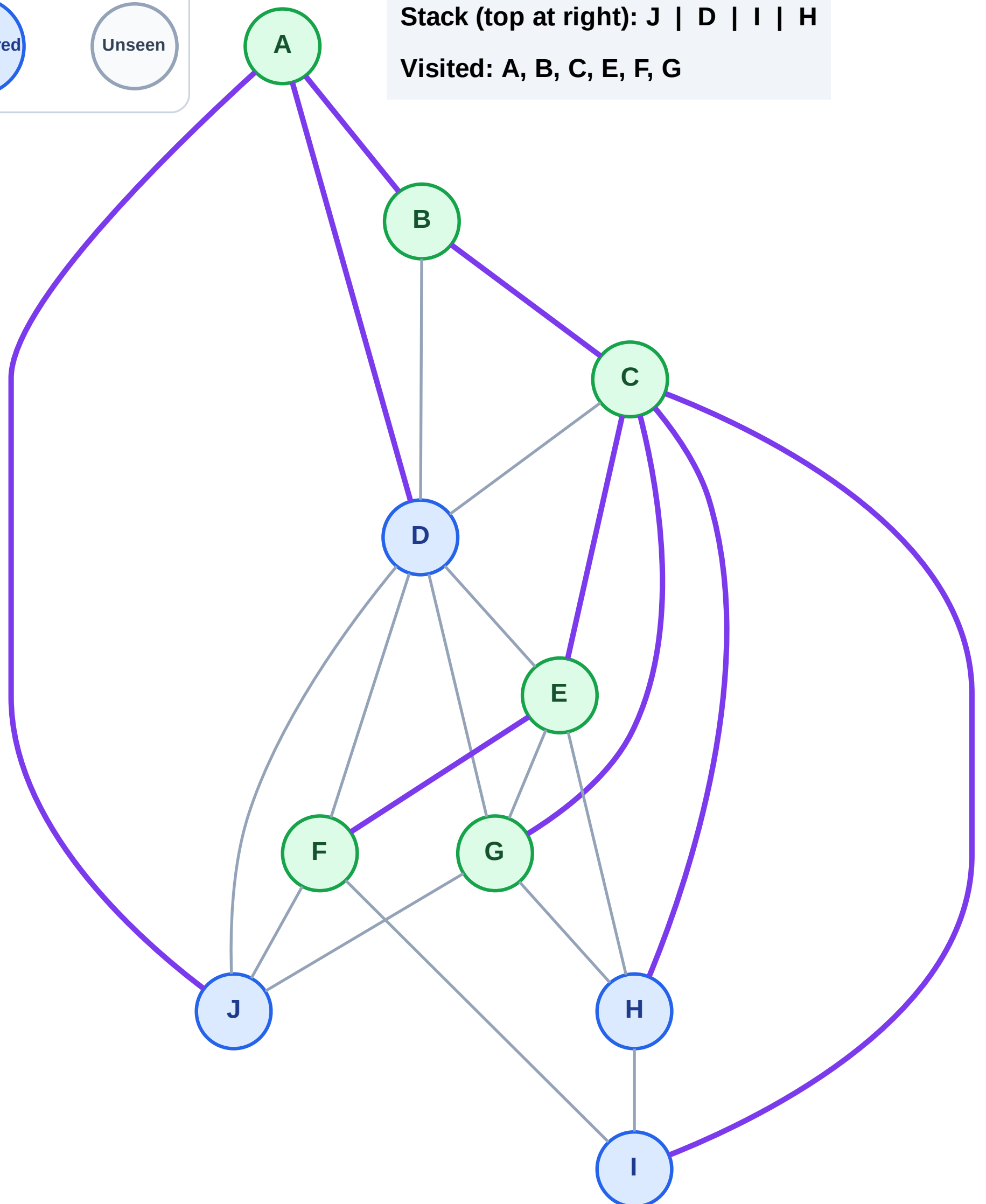
Step 13: No new neighbors from G

Legend



Stack (top at right): J | D | I | H

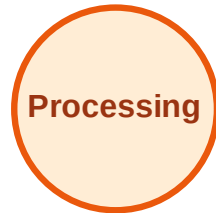
Visited: A, B, C, E, F, G



DFS Traversal

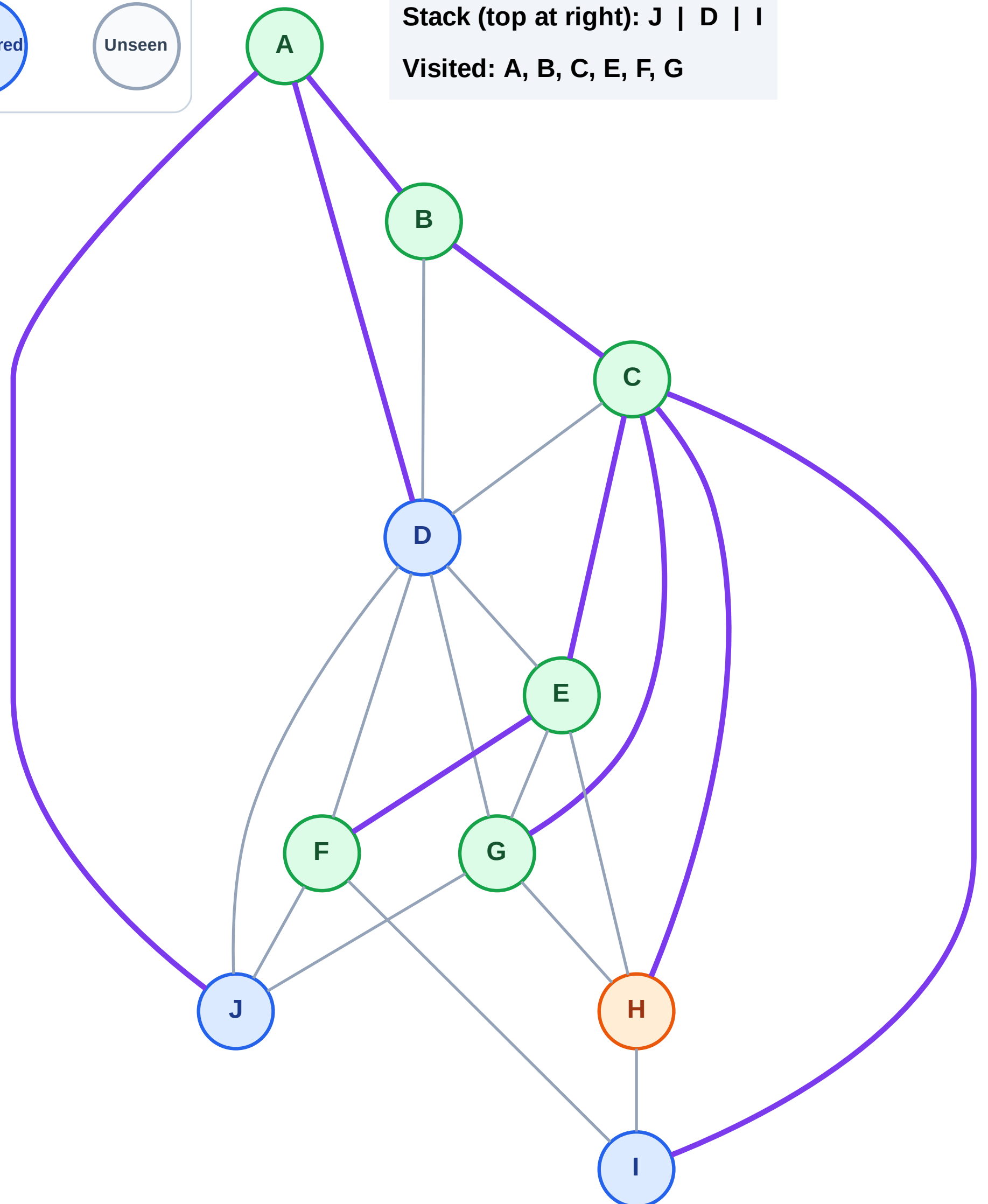
Step 14: Process node H

Legend



Stack (top at right): J | D | I

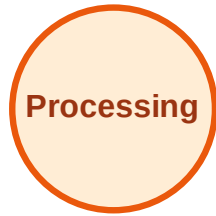
Visited: A, B, C, E, F, G



DFS Traversal

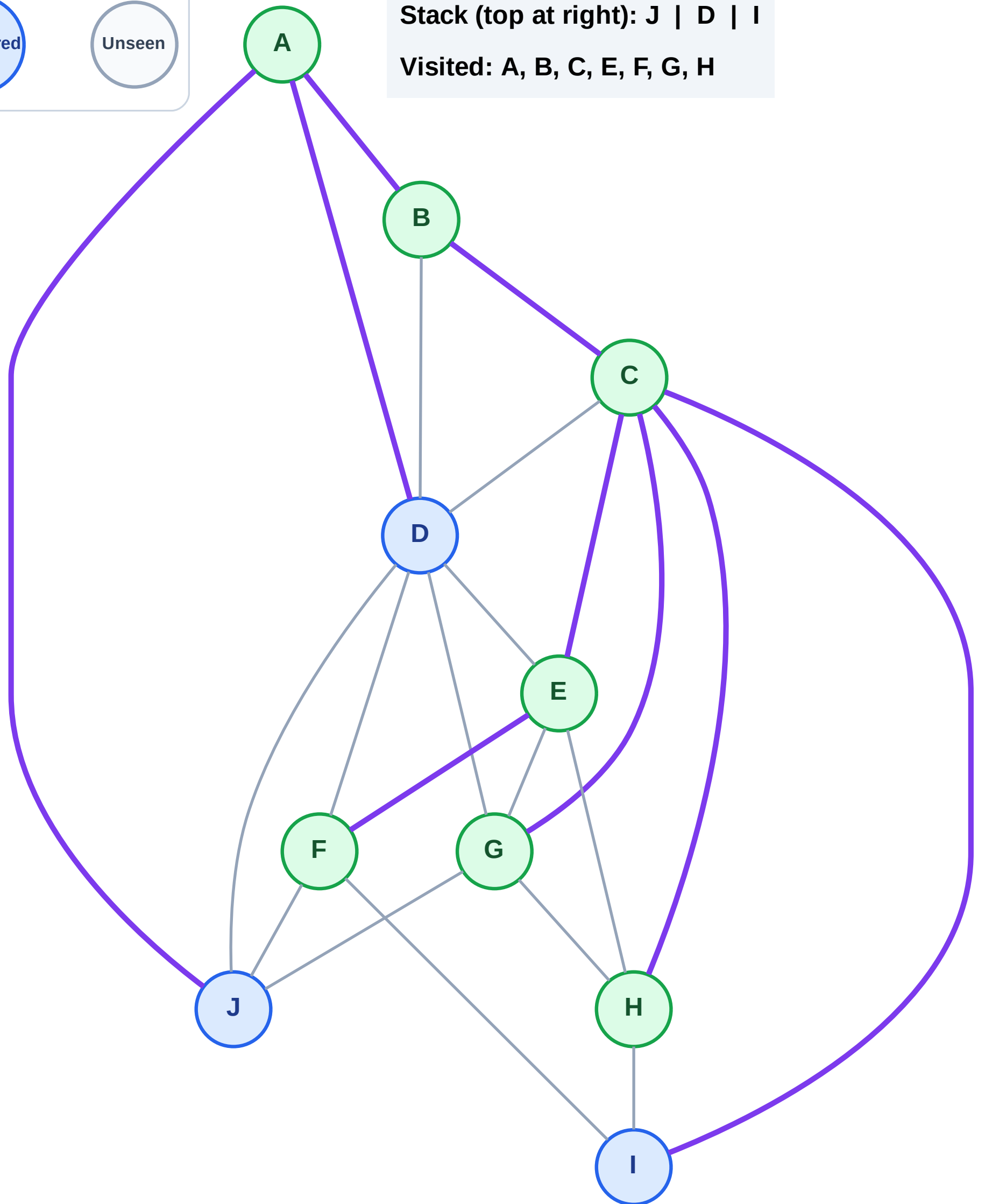
Step 15: No new neighbors from H

Legend



Stack (top at right): J | D | I

Visited: A, B, C, E, F, G, H



DFS Traversal

Step 16: Process node I

Legend

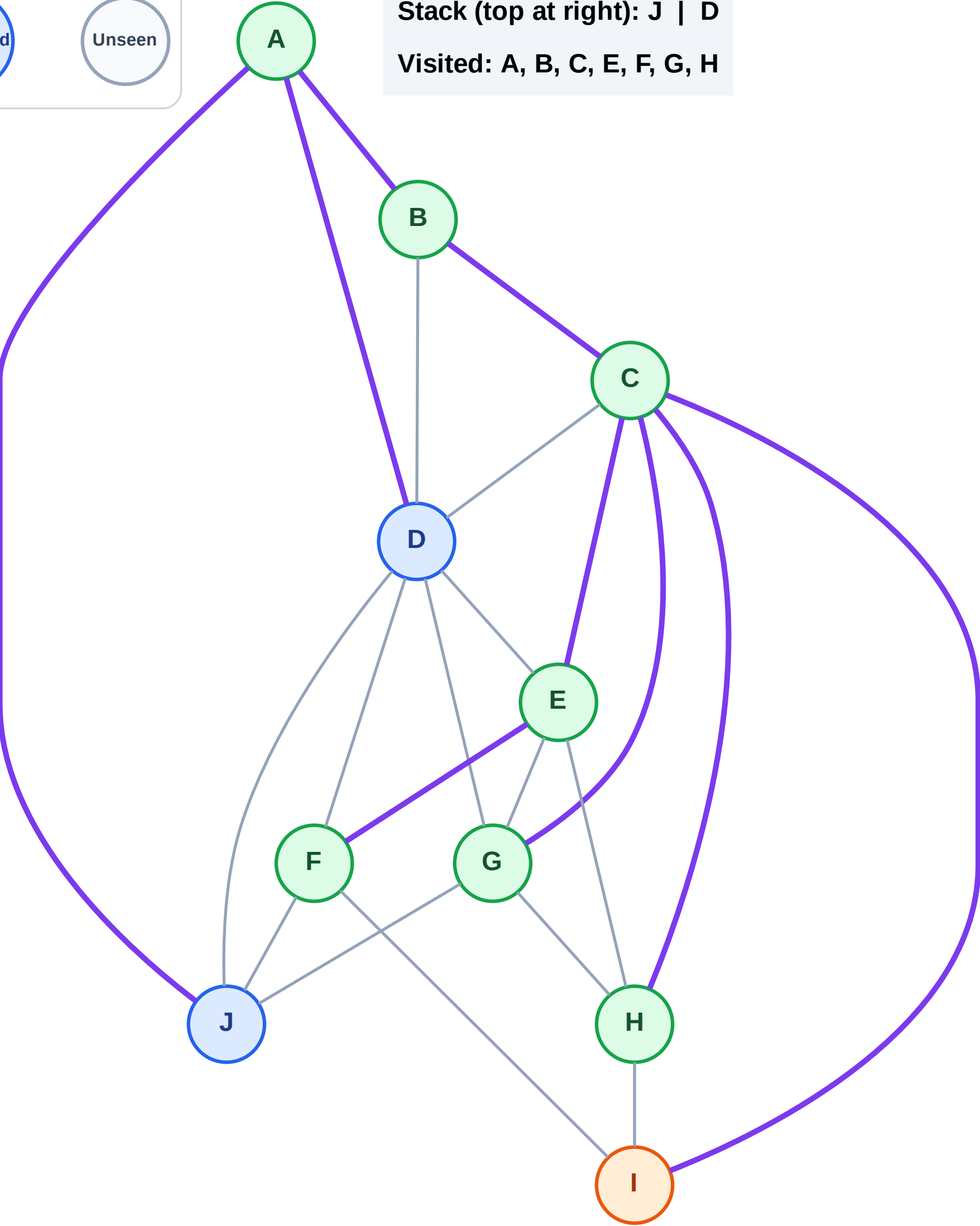
Complete

Processing

Discovered

Unseen

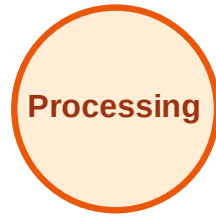
Stack (top at right): J | D
Visited: A, B, C, E, F, G, H



DFS Traversal

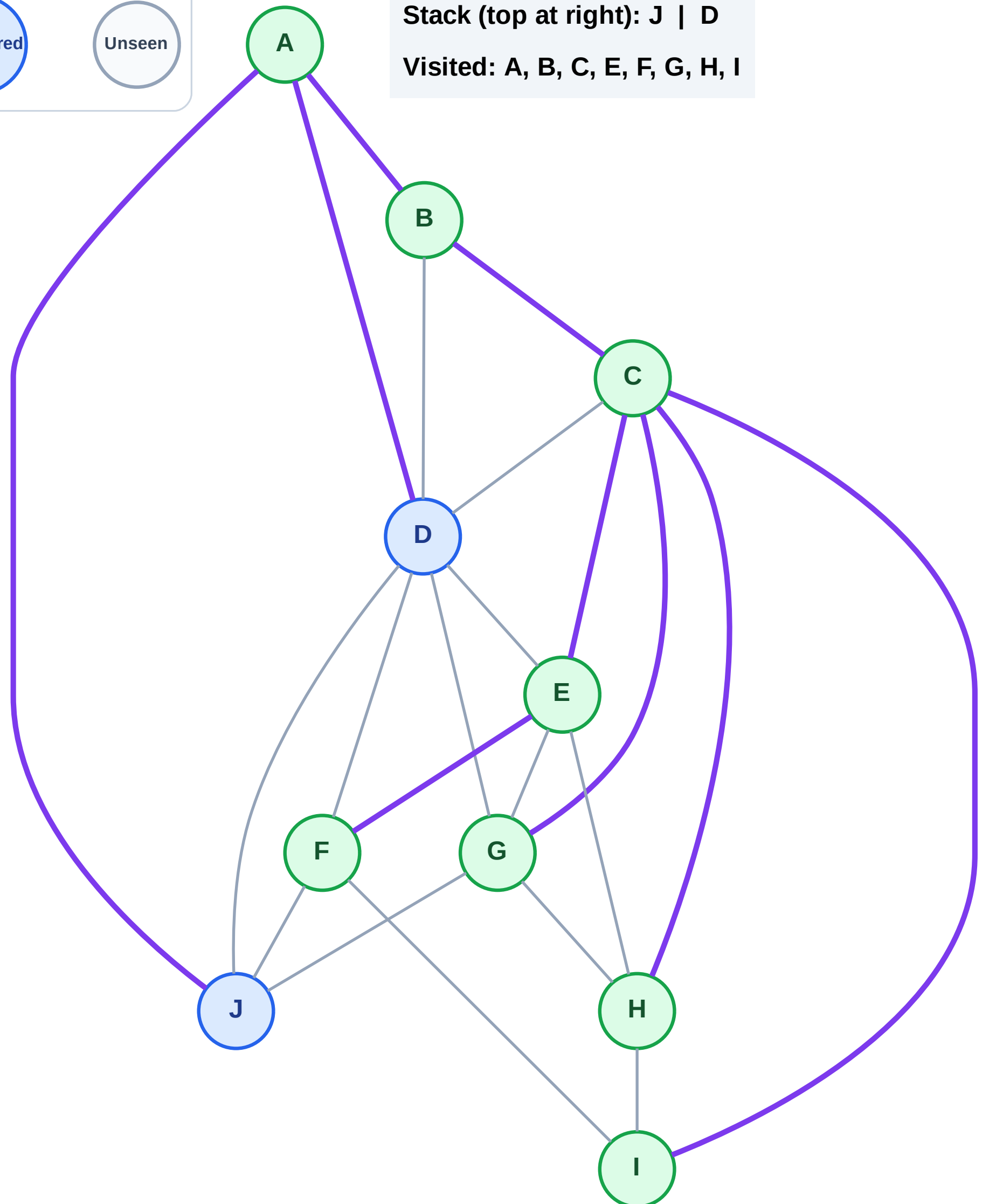
Step 17: No new neighbors from I

Legend



Stack (top at right): J | D

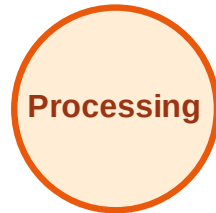
Visited: A, B, C, E, F, G, H, I



DFS Traversal

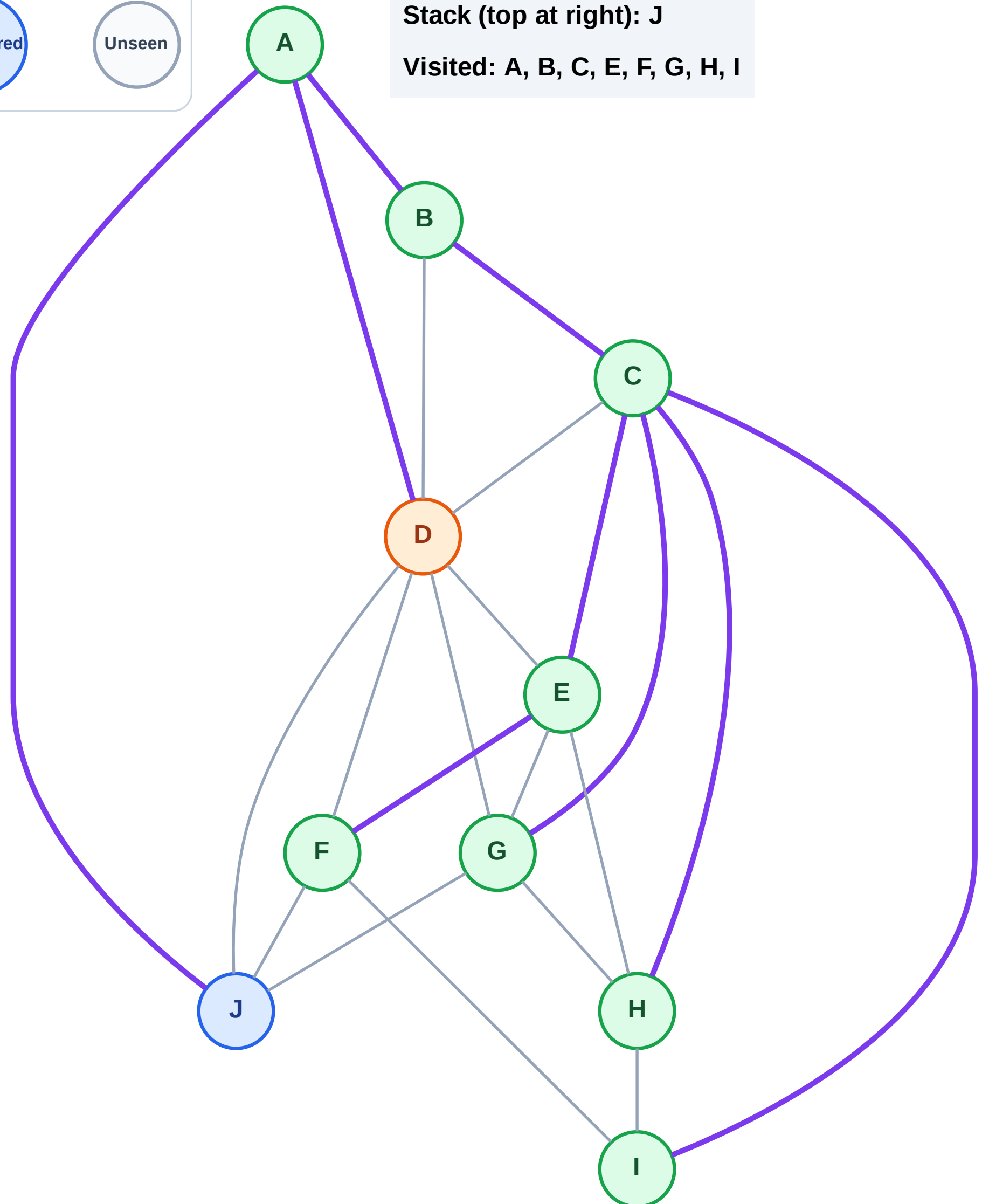
Step 18: Process node D

Legend



Stack (top at right): J

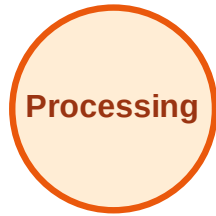
Visited: A, B, C, E, F, G, H, I



DFS Traversal

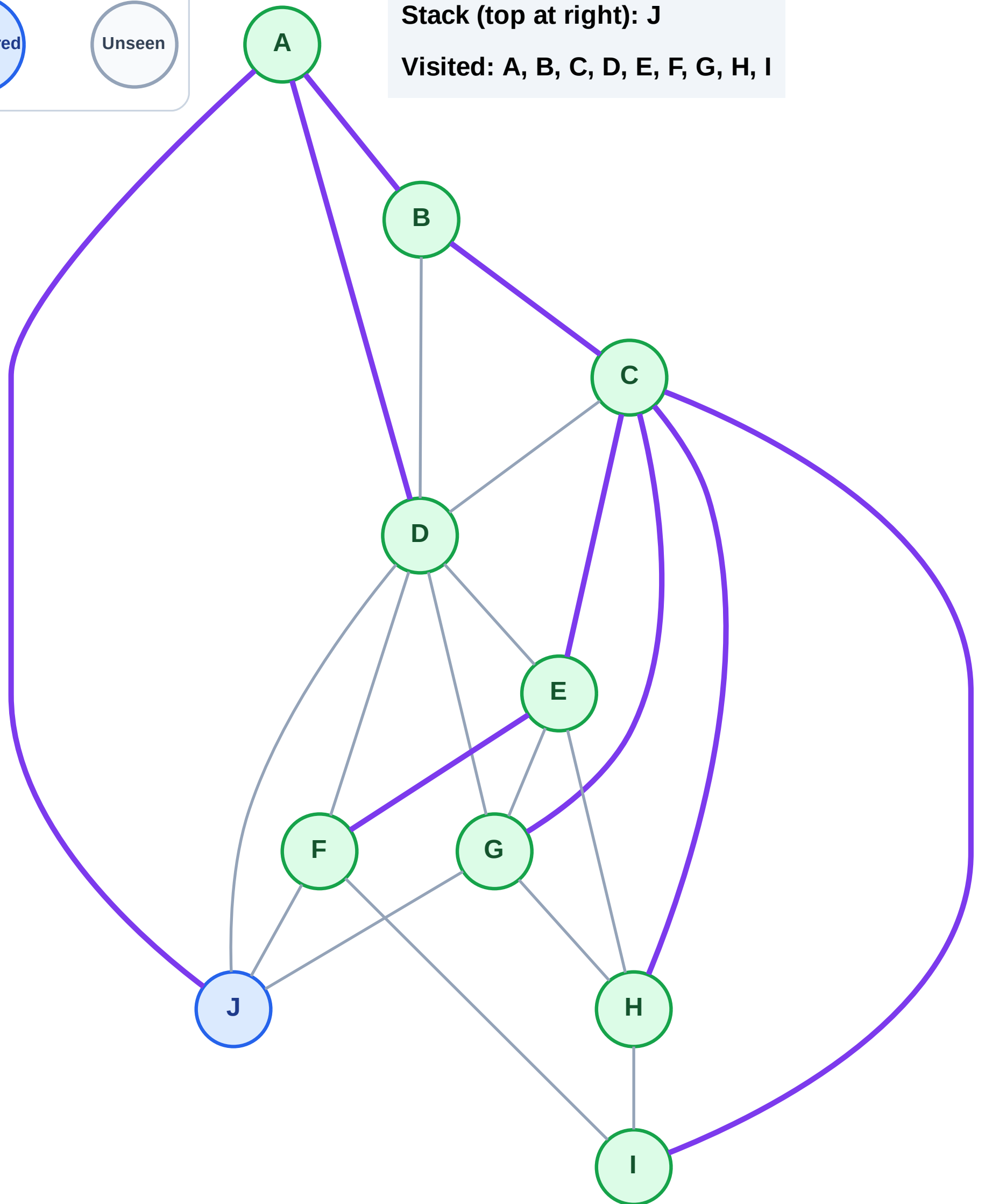
Step 19: No new neighbors from D

Legend



Stack (top at right): J

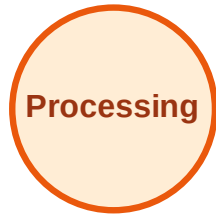
Visited: A, B, C, D, E, F, G, H, I



DFS Traversal

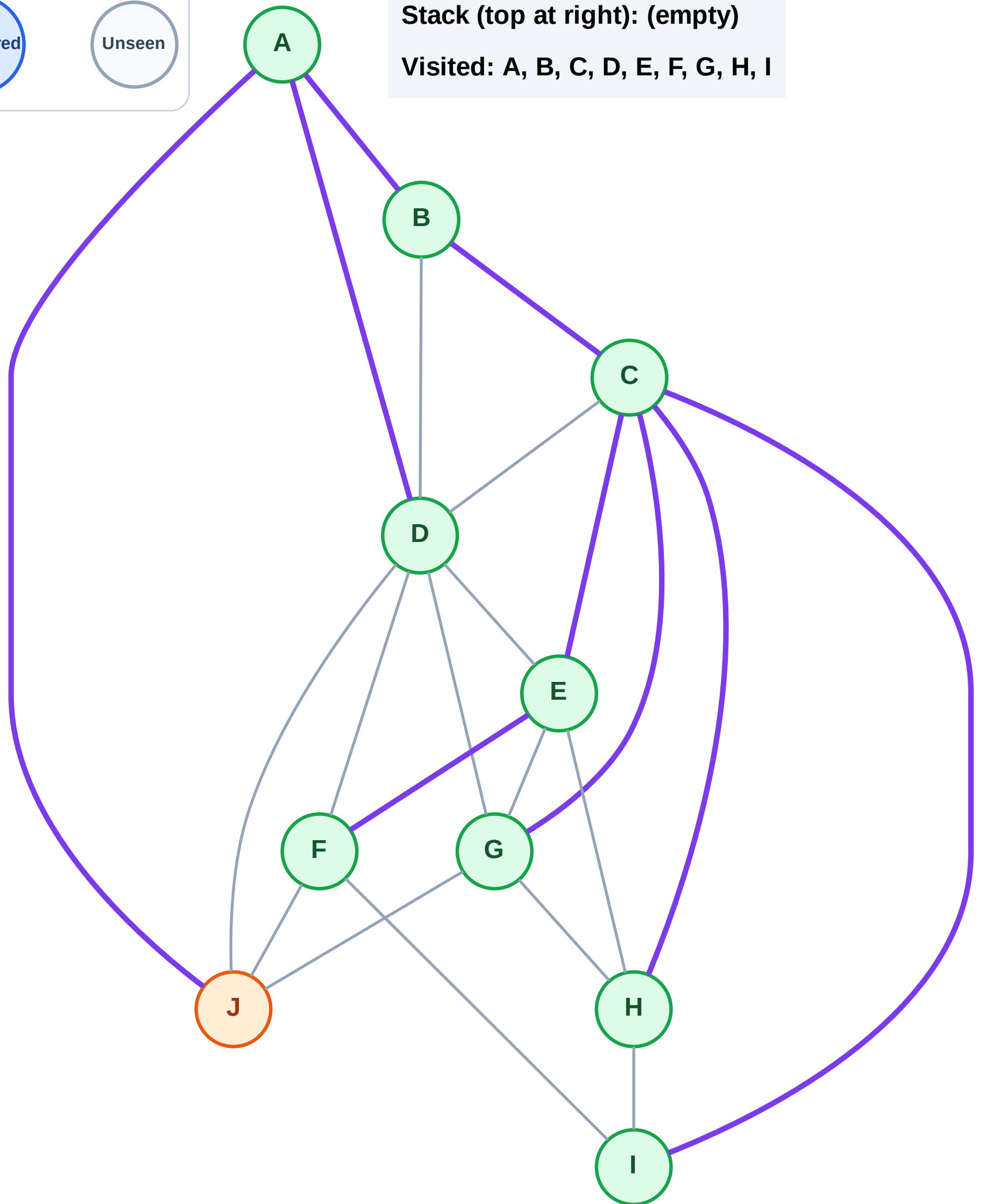
Step 20: Process node J

Legend



Stack (top at right): (empty)

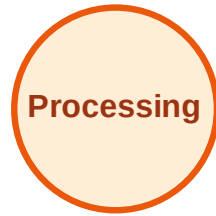
Visited: A, B, C, D, E, F, G, H, I



DFS Traversal

Step 21: No new neighbors from J

Legend



Stack (top at right): (empty)

Visited: A, B, C, D, E, F, G, H, I, J

